Perturbation expansion of the rate function for short-maturity variance options in local-stochastic volatility models: analyticity, explicit Heston solutions, and the Alòs connection

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April 4, 2026

Abstract

We establish the analyticity and derive an explicit perturbation expansion of the rate function governing the short-maturity asymptotics of options on realized variance in local-stochastic volatility (LSV) models. The rate function I_ρ, which arises from a constrained variational problem parametrized by the asset–volatility correlation ρ, is shown to be real-analytic in ρ on an interval $(-r, r)$ with $r > 0$, via the analytic implicit function theorem in Banach spaces applied to the Euler–Lagrange system. This yields the rigorous expansion

$$I_\rho = \sum_{k=0}^{N} I_k \, \rho^k + R_{N+1}(\rho), \qquad |R_{N+1}(\rho)| \le \frac{M \, |\rho|^{N+1}}{r - |\rho|},$$

where each coefficient I_k is explicit. The first-order correction $I_1 = -C[g_0, h_0]$ is a single integral over the known $\rho = 0$ optimal paths, computed via the envelope theorem, and the second-order correction I_2 reduces to a linear boundary value problem. For pure stochastic volatility models (constant local volatility), $I_1 = 0$ and the expansion is even in ρ, effectively doubling the convergence radius.

In the Heston model, we show that the $\rho = 0$ variational problem reduces to the classical Ermakov–Pinney equation, yielding the explicit optimal variance path $v(t) = V_0/(1 - p + p(t - 1)^2)$ and a closed-form expression for I_0 involving elementary functions. The parameter p is determined implicitly by the variance strike. We cross-validate this formula against the near-ATM asymptotics of Pirjol–Wang–Zhu and identify a structural prediction: I_0 saturates as the strike $K \to \infty$, implying that the variance option smile flattens in the far wing.

Finally, we establish a connection between our perturbation expansion and the Alòs decomposition of option prices. Both expansions are controlled by the same cross-correlation functional $C[g, h]$, but they operate at different points of the price distribution: the Alòs decomposition controls the bulk (moderate strikes), while our rate-function expansion controls the tail (extreme strikes). For constant local volatility, both independently predict $I_1 = 0$, providing a non-trivial cross-validation. The perturbation expansion is validated numerically by Monte Carlo simulation, achieving substantially greater accuracy than the existing bounds of Pirjol–Wang–Zhu for $|\rho| \le 0.7$.

Keywords: variance options, short-maturity asymptotics, large deviations, local-stochastic volatility, perturbation expansion, Ermakov–Pinney equation, Alòs decomposition
MSC 2020: 91G20, 60F10, 49K15, 34B15

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1 Introduction

Options on realized variance are derivative contracts whose payoff is linked to the annualized realized variance (or equivalently, the quadratic variation of the log-price) of an underlying asset over a fixed time period. Following the launch of CBOE S&P 500 Variance Futures (ticker: VA) in September 2024, these instruments have attracted renewed attention from both practitioners and researchers.

Short-maturity asymptotics provide a tractable analytical framework for pricing these options. Using large deviations theory, the leading-order short-maturity asymptotics of out-of-the-money (OTM) variance options can be expressed in terms of a rate function I_ρ, which arises as the solution of a two-dimensional variational problem involving the optimal fluctuation paths of the asset price and the volatility process.

Pirjol, Wang and Zhu Pirjol et al. [2025a] recently derived this variational problem for the general local-stochastic volatility (LSV) model and obtained two key results: (i) when the correlation ρ between asset returns and volatility innovations is zero, the rate function I_0 can be computed in closed form; (ii) for general ρ, they established the bounds

$$\frac{1}{1+|\rho|}\, I_0(S_0, V_0, K) \;\le\; I_\rho(S_0, V_0, K) \;\le\; \frac{1}{1-|\rho|}\, I_0(S_0, V_0, K), \tag{1}$$

together with an expansion around the at-the-money (ATM) point in powers of log-moneyness. However, the authors noted that the Euler–Lagrange equations for the correlated case "seem impossible to solve in closed form."

In this paper, we bridge the gap between the explicit $\rho = 0$ solution and the general correlated case by establishing a rigorous perturbation expansion of I_ρ around $\rho = 0$. Our contributions are as follows.

(i) **Analyticity.** We prove that I_ρ is real-analytic in ρ on an interval $(-r, r)$ with $r > 0$, via the analytic implicit function theorem in Banach spaces applied to the Euler–Lagrange system. This provides rigorous error bounds for the truncated expansion and identifies the structural obstructions to convergence (Theorem 19).

(ii) **Perturbation coefficients.** The first-order correction I_1 is given by a single integral over the known $\rho = 0$ optimal paths, computed via the envelope theorem (Theorem 6). The second-order correction I_2 is expressed in terms of the solution of a linear boundary value problem derived from the linearized Euler–Lagrange equations (Theorem 9).

(iii) **Explicit Heston solution.** We show that in the Heston model, the $\rho = 0$ variational problem reduces to the classical Ermakov–Pinney equation $u'' = -\lambda/u^3$, yielding a fully explicit optimal variance path and a closed-form expression for I_0 involving only elementary functions. We cross-validate against the near-ATM formula of Pirjol et al. [2025a] and identify far-wing saturation (Theorem 25).

(iv) **Alòs decomposition connection.** We establish that our perturbation expansion and the Alòs decomposition of option prices are controlled by the same cross-correlation functional $C[g, h]$, operating at different regimes of the price distribution. For constant local volatility, both independently predict $I_1 = 0$ (Theorem 33).

(v) **Numerical validation.** Monte Carlo simulation in the Tanh LSV model confirms the accuracy of the expansion for $|\rho| \le 0.7$, with the second-order approximation substantially outperforming the bounds (1).

Our approach is inspired by the near-ATM expansion technique developed in the companion paper Pirjol and Zhu [2025] for Asian options in LSV models. However, it is important to distinguish the two expansion strategies. In Pirjol and Zhu [2025], the perturbation parameter is the log-moneyness $x = \log(K/K_{\mathrm{ATM}})$, and the expansion is performed around the at-the-money point $x = 0$; this yields accurate approximations for near-ATM strikes at arbitrary correlation $\rho \in (-1, 1)$. In the present paper, we instead take ρ itself as the perturbation parameter and expand around the uncorrelated case $\rho = 0$, which provides useful approximations across the full range of strikes but requires $|\rho|$ not too large. The two approaches are therefore complementary: the log-moneyness expansion of Pirjol and Zhu [2025] is "horizontal" (valid near ATM for all ρ), while our correlation expansion is "vertical" (valid for all strikes at moderate ρ). In the overlap region—near-ATM strikes at moderate ρ—both expansions agree, as we verify in Section 4.5.

Related literature. The large deviations approach to short-maturity option pricing was initiated by Varadhan Varadhan [1967] and developed for stochastic volatility models by Robertson Robertson [2010] and Forde–Jacquier Forde and Jacquier [2011]. Variance options have been studied by Carr–Geman–Madan–Yor Carr et al. [2005] and more recently in the rough volatility setting by Lacombe–Muguruza–Stone Lacombe et al. [2021]. The Aloș decomposition of option prices under stochastic volatility was introduced in Aloòs [2006, 2012] and extended to Asian-type options by Aloș–Nualart–Pravosud Aloòs et al. [2024]. The Ermakov–Pinney equation Ermakov [1880], Pinney [1950] is a classical nonlinear ODE that arises in quantum mechanics, cosmology, and optimal control; to our knowledge, this is its first appearance in the context of variance option pricing. The wing asymptotics of implied volatility are controlled by the rate function through the work of Friz–Gerhold–Gulisashvili Friz et al. [2015] and Lee Lee [2004].

Organization. Section 2 introduces the model and the variational problem. Section 3 reviews the known $\rho = 0$ results. Section 4 presents the perturbation expansion with complete derivations. Section 5 establishes the analyticity of I_ρ and derives rigorous error bounds. Section 6 develops the explicit Heston solution via the Ermakov–Pinney equation. Section 7 establishes the connection with the Aloș decomposition. Section 8 provides numerical validation. Section 9 analyzes the boundary behavior of the expansion. Section 10 discusses practical implications and Section 11 concludes.

2 Model and variational problem

We consider the local-stochastic volatility model of Pirjol et al. [2025a]: under the risk-neutral measure $\mathbb{Q}$, the asset price S_t and variance process V_t satisfy

$$\frac{\mathrm{d}S_t}{S_t} = (r - q)\,\mathrm{d}t + \eta(S_t)\sqrt{V_t}\big(\rho\,\mathrm{d}Z_t + \sqrt{1-\rho^2}\,\mathrm{d}W_t\big), \tag{2}$$

$$\frac{\mathrm{d}V_t}{V_t} = \mu(V_t)\,\mathrm{d}t + \sigma(V_t)\,\mathrm{d}Z_t, \tag{3}$$

where W_t, Z_t are independent standard Brownian motions, $\rho \in (-1, 1)$ is the correlation parameter, and $\eta(\cdot), \mu(\cdot), \sigma(\cdot)$ satisfy the following regularity conditions.

Assumption 1 (Regularity)**.** The functions $\eta : (0,\infty) \to (0,\infty)$, $\mu : (0,\infty) \to \mathbb{R}$, and $\sigma : (0,\infty) \to (0,\infty)$ are real-analytic on $(0,\infty)$. There exist constants $c_\eta, C_\eta, c_\sigma, C_\sigma > 0$ such that

$$c_\eta \le \eta(S) \le C_\eta, \qquad c_\sigma \le \sigma(V) \le C_\sigma \tag{4}$$

for all S, V in compact subsets of $(0,\infty)$.

Assumption 2 (Strict positivity)**.** We assume $\eta(S) > 0$ for all $S > 0$, ensuring that the local volatility function is strictly positive and the functionals defined below are well-defined.

Assumption 3 (Non-degeneracy of the variational problem)**.** The infimum in the variational problem (7) below is attained at a unique minimizer $(g_0, h_0) \in H^1[0,1]^2$, and the bordered second variation (defined precisely in Section 5) is positive definite on the tangent space to the constraint manifold at (g_0, h_0).

The payoff of a variance call option with strike K and maturity T is

$$C(T) = e^{-rT}\,\mathbb{E}\left[\left(\frac{1}{T}\int_0^T V_s\,\eta^2(S_s)\,\mathrm{d}s - K\right)^+\right]. \tag{5}$$

The short-maturity asymptotics for OTM variance options is governed by the rate function

$$\lim_{T\to 0} T\log C(T) = -I_\rho(S_0, V_0, K), \tag{6}$$

where, by Theorem 4 of Pirjol et al. [2025a],

$$I_\rho(S_0, V_0, K) = \inf_{\substack{g(0)=\log S_0,\ h(0)=\log V_0 \\ \int_0^1 e^{h(t)}\eta^2(e^{g(t)})\,\mathrm{d}t = K}} \Lambda_\rho[g,h], \tag{7}$$

and the action functional is

$$\Lambda_\rho[g,h] = \frac{1}{2(1-\rho^2)}\int_0^1 \left(\frac{g'(t)}{\eta(e^{g(t)})\sqrt{e^{h(t)}}} - \frac{\rho\,h'(t)}{\sigma(e^{h(t)})}\right)^2 \mathrm{d}t + \frac{1}{2}\int_0^1 \left(\frac{h'(t)}{\sigma(e^{h(t)})}\right)^2 \mathrm{d}t. \tag{8}$$

The infimum is taken over absolutely continuous paths $g, h \in \mathrm{AC}[0,1]$ satisfying the boundary conditions and the integral constraint. The transversality conditions at $t = 1$ require $g'(1) = h'(1) = 0$.

2.1 Decomposition of the action functional

For the perturbation analysis, it is convenient to decompose Λ_ρ using three fundamental quadratic forms. Expanding the square in (8) and collecting terms, we obtain

$$\Lambda_\rho[g,h] = \frac{1}{2(1-\rho^2)}\big(A[g,h] + B[g,h] - 2\rho\,C[g,h]\big), \tag{9}$$

or equivalently, separating the three quadratic forms,

$$\Lambda_\rho[g,h] = \frac{1}{2(1-\rho^2)}\big(A[g,h] + B[g,h]\big) - \frac{\rho}{1-\rho^2}\,C[g,h], \tag{10}$$

where the three fundamental functionals are:

$$A[g,h] := \int_0^1 \frac{(g'(t))^2}{\eta^2(e^{g(t)})\,e^{h(t)}}\,\mathrm{d}t, \tag{11}$$

$$B[g,h] := \int_0^1 \frac{(h'(t))^2}{\sigma^2(e^{h(t)})}\,\mathrm{d}t, \tag{12}$$

$$C[g,h] := \int_0^1 \frac{g'(t)\,h'(t)}{\eta\big(e^{g(t)}\big)\,e^{h(t)/2}\,\sigma\big(e^{h(t)}\big)}\,\mathrm{d}t. \tag{13}$$

Definition 1 (Fundamental quadratic forms)**.** We refer to $A[g,h]$ as the *asset action*, $B[g,h]$ as the *volatility action*, and $C[g,h]$ as the *cross-correlation functional*. When evaluated at the optimal paths (g_0,h_0), we write $A_0 := A[g_0,h_0]$, $B_0 := B[g_0,h_0]$, and $C_0 := C[g_0,h_0]$.

Note that $\Lambda_0[g,h] = \frac{1}{2}(A[g,h]+B[g,h])$ and $I_0 = \frac{1}{2}(A_0+B_0)$.

Lemma 2 (Cauchy–Schwarz bound)**.** *For any $(g,h) \in \mathrm{AC}[0,1]^2$, the cross-correlation functional satisfies*

$$|C[g,h]| \le \sqrt{A[g,h]}\cdot\sqrt{B[g,h]} \le \frac{A[g,h]+B[g,h]}{2} = \Lambda_0[g,h]. \tag{14}$$

Equality in the first inequality holds if and only if $g'(t)/(\eta(e^{g(t)})e^{h(t)/2})$ and $h'(t)/\sigma(e^{h(t)})$ are proportional pointwise almost everywhere on $[0,1]$.

Proof. Apply the Cauchy–Schwarz inequality to the integral defining $C[g,h]$:

$$|C[g,h]|^2 = \left|\int_0^1 \underbrace{\frac{g'(t)}{\eta(e^{g(t)})e^{h(t)/2}}}_{=:u(t)}\cdot\underbrace{\frac{h'(t)}{\sigma(e^{h(t)})}}_{=:w(t)}\,\mathrm{d}t\right|^2 \le \int_0^1 u^2\,\mathrm{d}t\cdot\int_0^1 w^2\,\mathrm{d}t = A[g,h]\cdot B[g,h]. \tag{15}$$

The AM-GM inequality gives $\sqrt{AB}\le(A+B)/2$. □

Expanding $1/(1-\rho^2) = \sum_{n=0}^{\infty}\rho^{2n}$ gives the series representation

$$\Lambda_\rho[g,h] = \Lambda_0[g,h] - \rho\,C[g,h] + \rho^2\,\Lambda_0[g,h] - \rho^3\,C[g,h] + \cdots \tag{16}$$

valid for $|\rho|<1$. This formal expansion of the action functional at fixed paths is the starting point for the perturbation analysis. The key subtlety is that the *optimal paths* also vary with ρ, as we address carefully in the subsequent sections.

2.2 The constraint manifold

Define the constraint functional

$$\Phi[g,h] := \int_0^1 e^{h(t)}\,\eta^2\big(e^{g(t)}\big)\,\mathrm{d}t - K. \tag{17}$$

The constraint set is $\mathcal{C} := \{(g,h)\in H^1[0,1]^2 : g(0)=\log S_0,\ h(0)=\log V_0,\ \Phi[g,h]=0\}$. Under Assumption 2, the Fréchet derivative of Φ is

$$\delta\Phi[g,h]\cdot(\phi,\chi) = \int_0^1 e^{h(t)}\big[\chi(t)\,\eta^2(e^{g(t)}) + 2\eta(e^{g(t)})\eta'(e^{g(t)})e^{g(t)}\,\phi(t)\big]\,\mathrm{d}t. \tag{18}$$

For $K>0$ and $(g,h)\in\mathcal{C}$, the integrand contains the term $\chi(t)\,\eta^2(e^{g(t)})e^{h(t)}>0$, so $\delta\Phi$ is surjective. This ensures that $\mathcal{C}$ is a smooth Banach manifold.

3 Known results at $\rho = 0$

When $\rho = 0$, the variational problem (7) was solved in closed form by Pirjol, Wang and Zhu Pirjol et al. [2025a]. We summarize the key results needed for our expansion.

Proposition 3 (Proposition 6 and Remark 10 of Pirjol et al. [2025a])**.** *Suppose Assumptions 1–3 hold and $\eta(\cdot)$ is non-increasing. When $\rho = 0$, the rate function $I_0(S_0, V_0, K)$ is given by a one-dimensional optimization problem involving the rate function $J(V_0, z)$ for Asian options in a local volatility model with volatility $\sigma(\cdot)$, as specified in equations (3.6)–(3.8) of Pirjol et al. [2025a]. The optimal paths (g_0, h_0) that achieve the infimum in (7) at $\rho = 0$ are given explicitly in Remark 10 of Pirjol et al. [2025a].*

In the special case of constant local volatility $\eta(\cdot) \equiv \eta_0$ and constant volatility-of-volatility $\sigma(\cdot) \equiv \sigma_0$, the rate function simplifies considerably:

$$I_0(S_0, V_0, K) = \inf_{z>0} \left\{ F_c(z/K) + \frac{1}{\sigma_0^2} J_{\mathrm{BS}}\left(\frac{zK}{V_0}\right) \right\}, \tag{19}$$

where $J_{\mathrm{BS}}(k)$ is the rate function for Asian options in the Black–Scholes model (given in closed form by Proposition 12 of Pirjol and Zhu [2016]), and F_c is an explicit function depending on η_0 (see Section 4.1 of Pirjol et al. [2025a]).

3.1 Structure of the $\rho = 0$ optimal paths

The following structural properties of the $\rho = 0$ optimal paths are essential for the perturbation analysis.

Lemma 4 (Separation of asset and volatility paths at $\rho = 0$)**.** *When $\eta(\cdot) \equiv \eta_0$ is constant (pure stochastic volatility model), the $\rho = 0$ optimal asset path is identically constant: $g_0(t) \equiv \log S_0$ for all $t \in [0, 1]$. The entire variational problem reduces to optimizing over the volatility path h_0 alone:*

$$I_0 = \inf_{\substack{h(0)=\log V_0,\ h'(1)=0 \\ \int_0^1 \eta_0^2\, e^{h(t)}\, \mathrm{d}t = K}} \frac{1}{2} B[h], \tag{20}$$

where $B[h] := \int_0^1 (h'(t))^2 / \sigma^2(e^{h(t)})\, \mathrm{d}t$.

Proof. When $\eta \equiv \eta_0$, the functional $A[g, h] = (1/\eta_0^2) \int_0^1 (g')^2 e^{-h}\, \mathrm{d}t$ is minimized by $g_0 \equiv \log S_0$ (a constant) independently of h, since $A \geq 0$ with equality iff $g' \equiv 0$. The constraint $\int_0^1 \eta_0^2 e^h\, \mathrm{d}t = K$ depends only on h, so the asset and volatility optimizations separate. □

Lemma 5 (Transversality and monotonicity)**.** *Under the conditions of Proposition 3, the $\rho = 0$ optimal volatility path h_0 satisfies:*

(a) $h_0'(1) = 0$ (transversality condition).

(b) For $K > V_0 \eta_0^2$ (OTM variance call), h_0 is strictly increasing on $[0, 1)$ and achieves its maximum at $t = 1$.

(c) For $K < V_0 \eta_0^2$ (OTM variance put), h_0 is strictly decreasing on $[0, 1)$ and achieves its minimum at $t = 1$.

Proof. Part (a) is the standard transversality condition for free endpoint at $t = 1$. Parts (b) and (c) follow from the structure of the Euler–Lagrange equation: the optimal path must satisfy $\int_0^1 \eta_0^2 e^h \, \mathrm{d}t = K$, and the path that minimizes $B[h]$ subject to this constraint is monotone, since any non-monotone path can be replaced by a rearrangement with lower action (by the Polya–Szegő inequality applied to the appropriate functional). □

4 Main result: perturbation expansion

4.1 First-order correction

Our first main result establishes the first-order correction to the rate function.

Theorem 6 (First-order perturbation)**.** *Under Assumptions 1–3, the rate function admits the expansion*

$$I_\rho = I_0 - \rho\, C[g_0, h_0] + O(\rho^2), \tag{21}$$

where $C[g_0, h_0]$ is the cross-correlation integral (13) *evaluated on the $\rho = 0$ optimal paths.*

Equivalently,

$$I_1 := \left.\frac{\mathrm{d}I_\rho}{\mathrm{d}\rho}\right|_{\rho=0} = -\int_0^1 \frac{g_0'(t)\, h_0'(t)}{\eta\big(e^{g_0(t)}\big)\, e^{h_0(t)/2}\, \sigma\big(e^{h_0(t)}\big)} \, \mathrm{d}t = -C_0. \tag{22}$$

Proof. We apply the envelope theorem (Danskin's theorem) to the parametric optimization problem (7). Write $I_\rho = \inf_{(g,h)\in\mathcal{C}} \Lambda_\rho[g, h]$, where $\mathcal{C}$ denotes the constraint set. By the envelope theorem, assuming the optimizer is unique and smooth in ρ (guaranteed by Assumptions 3 and the implicit function theorem argument of Theorem 19 below),

$$\frac{\mathrm{d}I_\rho}{\mathrm{d}\rho} = \left.\frac{\partial \Lambda_\rho}{\partial \rho}\right|_{g=g^*(\rho),\, h=h^*(\rho)}, \tag{23}$$

where $(g^*(\rho), h^*(\rho))$ is the optimizer at correlation ρ.

From the decomposition (9)–(10), we compute the partial derivative with respect to ρ at fixed (g, h):

$$\frac{\partial \Lambda_\rho}{\partial \rho} = \frac{\rho}{(1-\rho^2)^2}\big(A[g,h] + B[g,h]\big) - \frac{1+\rho^2}{(1-\rho^2)^2}\, C[g,h]. \tag{24}$$

We verify this by differentiating (9):

$$\frac{\partial}{\partial \rho}\left[\frac{1}{2(1-\rho^2)}(A + B - 2\rho C)\right] = \frac{\rho}{(1-\rho^2)^2}(A + B - 2\rho C) + \frac{1}{2(1-\rho^2)}(-2C) \tag{25}$$

$$= \frac{\rho(A+B)}{(1-\rho^2)^2} - \frac{2\rho^2 C}{(1-\rho^2)^2} - \frac{C}{1-\rho^2} \tag{26}$$

$$= \frac{\rho(A+B)}{(1-\rho^2)^2} - \frac{(1+\rho^2)C}{(1-\rho^2)^2}. \tag{27}$$

Evaluating at $\rho = 0$ with $(g, h) = (g_0, h_0)$:

$$\left.\frac{\partial \Lambda_\rho}{\partial \rho}\right|_{\rho=0} = 0 \cdot (A_0 + B_0) - 1 \cdot C_0 = -C_0, \tag{28}$$

which gives (22). □

Remark 7 (Sign of I_1 and consistency with PWZ bounds)**.** *By the Cauchy–Schwarz inequality of Lemma 2:*

$$|I_1| = |C_0| \le \sqrt{A_0} \cdot \sqrt{B_0} \le \frac{A_0 + B_0}{2} = I_0. \tag{29}$$

This ensures $|I_1| \le I_0$, which is consistent with the bounds (1) at first order in ρ. Expanding those bounds gives $(1 - |\rho|)I_0 \le I_\rho \le (1 + |\rho|)I_0 + O(\rho^2)$, which requires $|I_1| \le I_0$.

Remark 8 (Financial interpretation)**.** *The first-order correction $I_1 = -C_0$ measures the correlation between the optimal asset-price fluctuation $g_0'(t)/(\eta\sqrt{v})$ and the optimal volatility fluctuation $h_0'(t)/\sigma(v)$ along the $\rho = 0$ most-likely path. When these fluctuations are positively correlated ($C_0 > 0$), introducing a positive ρ reduces the rate function (making OTM options less suppressed), which is the expected financial effect: positive asset-volatility correlation increases the probability of large variance realizations when the asset price moves with the volatility.*

4.2 Second-order correction

For the second-order correction, we need the first-order perturbation of the optimal paths. Define the Lagrangian

$$\mathcal{L}_\rho[g, h; \lambda] := \Lambda_\rho[g, h] + \lambda \left(\int_0^1 e^{h(t)} \eta^2(e^{g(t)})\, \mathrm{d}t - K \right), \tag{30}$$

where λ is the Lagrange multiplier. At $\rho = 0$, the optimal triple (g_0, h_0, λ_0) satisfies the Euler–Lagrange equations:

$$\frac{\mathrm{d}}{\mathrm{d}t}\left[\frac{g_0'(t)}{\eta^2(e^{g_0})\, e^{h_0}} \right] = -\frac{(g_0')^2\, \eta'(e^{g_0})\, e^{g_0}}{\eta^3(e^{g_0})\, e^{h_0}} + 2\lambda_0\, e^{h_0}\, \eta(e^{g_0})\, \eta'(e^{g_0})\, e^{g_0}, \tag{31}$$

$$\frac{\mathrm{d}}{\mathrm{d}t}\left[\frac{h_0'(t)}{\sigma^2(e^{h_0})} \right] = -\frac{(g_0')^2}{2\,\eta^2(e^{g_0})\, e^{h_0}} - \frac{(h_0')^2\, \sigma'(e^{h_0})\, e^{h_0}}{\sigma^3(e^{h_0})} + \lambda_0\, e^{h_0}\, \eta^2(e^{g_0}). \tag{32}$$

Now expand $g^*(\rho) = g_0 + \rho\, g_1 + \rho^2 g_2 + \cdots$ and $h^*(\rho) = h_0 + \rho\, h_1 + \rho^2 h_2 + \cdots$ where (g_1, h_1) satisfies the linearized constraint

$$\int_0^1 e^{h_0(t)} \Big[h_1(t)\, \eta^2(e^{g_0(t)}) + 2\eta(e^{g_0(t)})\eta'(e^{g_0(t)})\, e^{g_0(t)}\, g_1(t) \Big]\, \mathrm{d}t = 0, \tag{33}$$

with $g_1(0) = h_1(0) = 0$ (since the initial conditions are independent of ρ) and the transversality conditions $g_1'(1) = h_1'(1) = 0$.

The pair (g_1, h_1) is determined by the stationarity of Λ_ρ at first order in ρ, which yields a *linear* boundary value problem. Writing $\delta^2\mathcal{L}_0$ for the second variation of the Lagrangian at (g_0, h_0, λ_0) and δC for the first variation of the cross-correlation functional (13), the linearized equations are

$$\delta^2\mathcal{L}_0[(g_1, h_1); (\phi, \chi)] = \delta C[g_0, h_0; (\phi, \chi)] \qquad \text{for all admissible } (\phi, \chi). \tag{34}$$

This is a fourth-order linear BVP in (g_1, h_1).

Theorem 9 (Second-order perturbation)**.** *Under the assumptions of Theorem 6, the second-order correction to the rate function is*

$$I_2 := \frac{1}{2} \frac{\mathrm{d}^2 I_\rho}{\mathrm{d}\rho^2}\bigg|_{\rho=0} = I_0 - \frac{1}{2}\, \delta C[g_0, h_0] \cdot (g_1, h_1), \tag{35}$$

where (g_1, h_1) is the solution of the linearized system (34)*, and $\delta C[g_0, h_0] \cdot (g_1, h_1)$ denotes the first variation of C at (g_0, h_0) in the direction (g_1, h_1):*

$$\delta C \cdot (g_1, h_1) = \frac{\mathrm{d}}{\mathrm{d}\epsilon}\bigg|_{\epsilon=0} C[g_0 + \epsilon g_1, h_0 + \epsilon h_1]. \tag{36}$$

Combining with Theorem 6, the rate function through second order is:

$$I_\rho = I_0 - \rho\, C_0 + \rho^2\big(I_0 - \tfrac{1}{2}\,\delta C \cdot (g_1, h_1)\big) + O(\rho^3). \tag{37}$$

Remark 10 (Solvability of the linearized BVP)**.** *The linearized system* (34) *is a fourth-order elliptic boundary value problem for (g_1, h_1). Its solvability follows from the Fredholm alternative: a solution exists if and only if the right-hand side $\delta C[g_0, h_0; \cdot]$ is orthogonal to the kernel of the adjoint of $\delta^2\mathcal{L}_0$. By Assumption 3, the second variation $\delta^2\mathcal{L}_0$ is positive definite on the tangent space to the constraint manifold, so its kernel is trivial, and the Fredholm alternative guarantees existence and uniqueness of (g_1, h_1).*

Proof. We compute $\mathrm{d}^2 I_\rho/\mathrm{d}\rho^2$ at $\rho = 0$ using the decomposition (16). Substituting the path expansion $g^* = g_0 + \rho g_1 + \rho^2 g_2 + \cdots$ into $I_\rho = \Lambda_\rho[g^*, h^*]$ and collecting terms at each order in ρ:

Order ρ^0: $I_0 = \Lambda_0[g_0, h_0]$.

Order ρ^1: The $O(\rho)$ terms are

$$I_1 = \delta\Lambda_0[g_0, h_0] \cdot (g_1, h_1) - C[g_0, h_0]. \tag{38}$$

Since (g_0, h_0, λ_0) satisfies the Euler–Lagrange equations for Λ_0 with the constraint, we have $\delta\mathcal{L}_0 \cdot (\phi, \chi) = 0$ for all admissible (ϕ, χ), so in particular $\delta\Lambda_0 \cdot (g_1, h_1) = -\lambda_0\, \delta\Phi \cdot (g_1, h_1) = 0$ where the last equality uses the linearized constraint (33). This confirms $I_1 = -C_0$.

Order ρ^2: Collecting all $O(\rho^2)$ terms:

$$2I_2 = \underbrace{\delta\mathcal{L}_0 \cdot (g_2, h_2)}_{=-\lambda_0\,\delta\Phi\cdot(g_2,h_2)} + \delta^2\Lambda_0[(g_1, h_1), (g_1, h_1)] - 2\,\delta C[g_0, h_0] \cdot (g_1, h_1) + 2\Lambda_0[g_0, h_0]. \tag{39}$$

The first term requires the constraint at $O(\rho^2)$:

$$\delta\Phi \cdot (g_2, h_2) + \tfrac{1}{2}\delta^2\Phi[(g_1, h_1), (g_1, h_1)] = 0, \tag{40}$$

so $\delta\mathcal{L}_0 \cdot (g_2, h_2) = \frac{\lambda_0}{2}\delta^2\Phi[(g_1, h_1)]$. Setting $(\phi, \chi) = (g_1, h_1)$ in the linearized system (34):

$$\delta^2\mathcal{L}_0[(g_1, h_1), (g_1, h_1)] = \delta C \cdot (g_1, h_1). \tag{41}$$

Since $\delta^2\mathcal{L}_0 = \delta^2\Lambda_0 + \lambda_0\, \delta^2\Phi$, we get

$$\delta^2\Lambda_0[(g_1, h_1)] + \lambda_0\, \delta^2\Phi[(g_1, h_1)] = \delta C \cdot (g_1, h_1). \tag{42}$$

Substituting into (39) and using the relations above:

$$2I_2 = \tfrac{\lambda_0}{2}\delta^2\Phi + \big(\delta C \cdot (g_1, h_1) - \lambda_0\, \delta^2\Phi\big) - 2\,\delta C \cdot (g_1, h_1) + 2I_0 \tag{43}$$

$$= -\tfrac{\lambda_0}{2}\delta^2\Phi - \delta C \cdot (g_1, h_1) + 2I_0. \tag{44}$$

The boundary terms from the second variation of the Lagrangian, evaluated using $\frac{\lambda_0}{2}\delta^2\Phi[(g_1, h_1)]$, cancel by the transversality conditions, yielding

$$2I_2 = 2I_0 - \delta C \cdot (g_1, h_1), \tag{45}$$

which gives (35). □

4.3 Higher-order corrections

The perturbation procedure extends to arbitrary order. The n-th order correction I_n involves:

(a) The path corrections $(g_1, h_1), \ldots, (g_{n-1}, h_{n-1})$, each determined by a linear BVP with increasing complexity.

(b) The derivatives $\partial^k \Lambda_\rho / \partial \rho^k$ at $\rho = 0$, which involve combinatorial sums of the functionals A, B, C evaluated on the path corrections.

Proposition 11 (General term structure)**.** *For $n \geq 1$, the n-th order correction has the form*

$$I_n = a_n \, I_0 + b_n \, C_0 + \textit{terms involving } (g_1, h_1), \ldots, (g_{\lfloor n/2 \rfloor}, h_{\lfloor n/2 \rfloor}), \tag{46}$$

where a_n, b_n are universal constants arising from the expansion of $1/(1-\rho^2)$. Specifically, $a_n = 1$ for n even and $a_n = 0$ for n odd, while $b_n = -1$ for n odd and $b_n = 0$ for n even (to leading order, before path corrections).

Proof sketch. This follows from the structure of the series (16): the ρ^{2k} terms contribute Λ_0 and the ρ^{2k+1} terms contribute $-C$. The path corrections enter at second order and modify the coefficients. □

4.4 Explicit formulas for constant coefficients

The perturbation expansion takes a particularly explicit form when the local volatility and volatility-of-volatility are constants: $\eta(\cdot) \equiv \eta_0$ and $\sigma(\cdot) \equiv \sigma_0$.

Corollary 12 (Constant coefficient case)**.** *When $\eta(\cdot) \equiv \eta_0$ and $\sigma(\cdot) \equiv \sigma_0$, the first-order correction is*

$$I_1 = -\frac{1}{\eta_0 \, \sigma_0} \int_0^1 g_0'(t) \, h_0'(t) \, e^{-h_0(t)/2} \, \mathrm{d}t, \tag{47}$$

where $g_0(t)$ and $h_0(t)$ are the $\rho = 0$ optimal paths.

The second-order correction I_2 admits a semi-explicit representation. When $\eta \equiv \eta_0$ and $\sigma \equiv \sigma_0$ are constant, the linearized BVP (34) for (g_1, h_1) becomes a system of the form

$$\begin{pmatrix} g_1'' \\ h_1'' \end{pmatrix} = \begin{pmatrix} \alpha(t) & \beta(t) \\ \gamma(t) & \delta(t) \end{pmatrix} \begin{pmatrix} g_1 \\ h_1 \end{pmatrix} + \begin{pmatrix} F_g(t) \\ F_h(t) \end{pmatrix}, \tag{48}$$

where the coefficient functions $\alpha, \beta, \gamma, \delta$ depend on g_0, h_0 and λ_0, and the forcing terms F_g, F_h arise from $\delta C[g_0, h_0; \cdot]$.

In the pure stochastic volatility sub-case ($\eta \equiv \eta_0$, $g_0 \equiv \log S_0$), the g_1-equation decouples:

$$g_1''(t) = \eta_0^2 \, e^{h_0(t)} \, \lambda_0 \, g_1(t) + \frac{\eta_0}{\sigma_0} \, h_0'(t) \, e^{h_0(t)/2}, \qquad g_1(0) = 0, \;\; g_1'(1) = 0. \tag{49}$$

Given the solution g_1, the second-order correction reduces to a single quadrature:

$$I_2 = I_0 - \frac{1}{2 \, \eta_0 \, \sigma_0} \int_0^1 h_0'(t) \, e^{-h_0(t)/2} \, g_1'(t) \, \mathrm{d}t. \tag{50}$$

Proof. The decoupling of the g_1-equation follows from $g_0 \equiv \log S_0$, which eliminates all cross-terms between g_1 and h_1 in the linearized system. The forcing term in (49) arises from $\delta C[g_0, h_0; (\phi, 0)]$ with $g_0' = 0$: differentiation gives

$$\delta_g C[g_0, h_0] \cdot \phi = \frac{1}{\eta_0 \sigma_0} \int_0^1 \phi'(t)\, h_0'(t)\, e^{-h_0(t)/2}\, \mathrm{d}t. \tag{51}$$

The formula (50) follows from (35) by computing $\delta C \cdot (g_1, h_1)$ with $h_1 = 0$ (which holds in the pure SV case since the h_1 equation has zero forcing when $g_0' = 0$). □

Corollary 13 (Stochastic volatility models: $I_1 = 0$)**.** *When $\eta(\cdot) \equiv \eta_0$ is constant, the $\rho = 0$ optimal asset path is $g_0(t) \equiv \log S_0$ for all $t \in [0, 1]$, so $g_0'(t) = 0$. Consequently, $C_0 = 0$ and $I_1 = 0$. The leading ρ-correction is at second order:*

$$I_\rho = I_0 + \rho^2\, I_2 + O(\rho^4) \qquad (\textit{constant } \eta). \tag{52}$$

In particular, for pure stochastic volatility models ($\eta \equiv 1$), I_ρ is an even function of ρ to all orders.

Proof. When η is constant, $g_0 \equiv \log S_0$ by Lemma 4, so $g_0' \equiv 0$ and $C_0 = 0$. The parity statement follows from the decomposition (9): under the reflection $\rho \to -\rho$, the optimal paths satisfy $(g^*(-\rho), h^*(-\rho)) = (-g^*(\rho) + 2\log S_0, h^*(\rho))$ (the asset path reflection) when η is constant, which preserves Λ_ρ and hence I_ρ. More precisely, A and B are invariant under $g \to 2\log S_0 - g$ (since $(g')^2 = (-(g - 2\log S_0)')^2$), and C changes sign, so $\Lambda_{-\rho}[g, h] = \Lambda_\rho[2\log S_0 - g, h]$ and therefore $I_{-\rho} = I_\rho$. This improves the error bound in (52): the $O(\rho^3)$ term is actually zero, and the next correction is $O(\rho^4)$. □

Remark 14. *Corollary 13 has an intuitive interpretation. In a stochastic volatility model, the $\rho = 0$ optimal fluctuation concentrates entirely in the variance process (h_0 path), with the asset price remaining at its spot value. There is no asset-price fluctuation to correlate with the volatility fluctuation, so the first-order effect of introducing correlation vanishes. The even-in-ρ property also explains why stochastic volatility models are less sensitive to the sign of ρ for variance options (unlike equity options, where the skew changes sign with ρ).*

For the general LSV model with non-constant η, the optimal asset path g_0 is non-trivial even at $\rho = 0$, because the local volatility function couples the asset dynamics to the variance constraint. In this case $I_1 \neq 0$ generically.

4.5 Near-ATM limit of the perturbation expansion

We verify our expansion against the near-ATM expansion of Proposition 19 in Pirjol et al. [2025a]. Near the ATM point $K = V_0\eta_0^2$, the rate function is $O(x^2)$ where $x = \log(K/(V_0\eta_0^2))$. Both the $\rho = 0$ rate function and the first-order correction vanish at $x = 0$ and admit expansions in powers of x.

From Proposition 19 of Pirjol et al. [2025a], the leading term of the rate function at general ρ is

$$I_\rho = \frac{3}{2(\sigma_0^2 + 4\rho\sigma_0\eta_1\sqrt{V_0} + 4\eta_1^2 V_0)}\, x^2 + O(x^3), \tag{53}$$

where $\eta_1 = \eta'(S_0)\, S_0$ is the local volatility elasticity. Expanding in ρ:

$$I_\rho = \frac{3}{2(\sigma_0^2 + 4\eta_1^2 V_0)}\, x^2 - \rho\, \frac{6\sigma_0\eta_1\sqrt{V_0}}{(\sigma_0^2 + 4\eta_1^2 V_0)^2}\, x^2 + O(\rho^2 x^2, x^3). \tag{54}$$

Proposition 15 (Near-ATM consistency)**.** *The near-ATM limit of the first-order correction I_1 from Theorem 6 agrees with the $O(\rho)$ coefficient in* (53)*:*

$$\lim_{x\to 0}\frac{I_1}{x^2} = -\frac{6\sigma_0\eta_1\sqrt{V_0}}{(\sigma_0^2+4\eta_1^2V_0)^2}. \tag{55}$$

In particular, when $\eta_1 = 0$ (constant local volatility), the near-ATM limit confirms $I_1 = 0$.

Proof. This follows from the computation of $C[g_0, h_0]$ in the near-ATM regime, where the optimal paths are perturbations of the constant paths $g_0 = \log S_0$, $h_0 = \log V_0$. The detailed computation is given in Section 8. □

5 Analyticity and convergence

In this section, we establish that the perturbation expansion (**??**) is not merely formal but represents a convergent Taylor series in a neighborhood of $\rho = 0$. This provides rigorous error bounds and identifies the structural obstructions to convergence.

5.1 The Euler–Lagrange map

We formulate the Euler–Lagrange system as a map between Banach spaces. Define

$$\mathcal{X} := \{(g,h) \in H^1[0,1]^2 : g(0) = \log S_0,\ h(0) = \log V_0\} \times \mathbb{R}, \tag{56}$$

and let $\mathcal{Y} := (H^{-1}[0,1])^2 \times \mathbb{R} \times \mathbb{R}^2$, where the last factor accounts for the transversality conditions. Define the Euler–Lagrange map $F : \mathcal{X} \times (-1,1) \to \mathcal{Y}$ by

$$F(g,h,\lambda;\rho) := \begin{pmatrix} \text{E-L equation for } g \text{ (from } \mathcal{L}_\rho) \\ \text{E-L equation for } h \text{ (from } \mathcal{L}_\rho) \\ \Phi[g,h] \\ g'(1) \\ h'(1) \end{pmatrix}. \tag{57}$$

The Euler–Lagrange equations for g and h are the weak-form equations obtained by requiring $\delta\mathcal{L}_\rho/\delta g = 0$ and $\delta\mathcal{L}_\rho/\delta h = 0$. By construction, (g_0, h_0, λ_0) satisfies $F(g_0, h_0, \lambda_0; 0) = 0$.

5.2 Analyticity of the action functional

Lemma 16 (Analyticity of Λ_ρ in ρ)**.** *For any fixed $(g,h) \in \mathcal{X}$, the map $\rho \mapsto \Lambda_\rho[g,h]$ is real-analytic on $(-1,1)$.*

Proof. The functionals $A[g,h]$, $B[g,h]$, and $C[g,h]$ are independent of ρ. The ρ-dependence enters only through the factor $1/(2(1-\rho^2))$ in the decomposition (9), which is analytic on $(-1,1)$ (the nearest singularities of $1/(1-\rho^2)$ are at $\rho = \pm 1$). Therefore $\Lambda_\rho[g,h]$ is a linear combination of ρ-independent functionals with analytic coefficients. □

Lemma 17 (Joint analyticity of F)**.** *Under Assumption 1, the Euler–Lagrange map F is real-analytic as a map $\mathcal{X} \times (-1,1) \to \mathcal{Y}$.*

Proof. The Euler–Lagrange equations involve:

(a) The functions η, σ, and their derivatives, which are analytic by Assumption 1.

(b) The factor $1/(1-\rho^2)$ and $\rho/(1-\rho^2)$, which are analytic in ρ on $(-1,1)$.

(c) Compositions $\eta(e^g)$, $\sigma(e^h)$, which are analytic in (g,h) by the chain rule for analytic functions.

(d) Standard integral functionals of (g,h,g',h'), which are analytic on H^1 under the regularity assumptions.

The joint analyticity follows from the fact that the composition, sum, and product of analytic maps between Banach spaces are analytic (see, e.g., [Bonnans and Shapiro, 2000, Chapter 4]). □

5.3 Invertibility of the bordered operator

The Fréchet derivative $D_{(g,h,\lambda)}F$ at $(g_0,h_0,\lambda_0;0)$ has the bordered structure

$$D_{(g,h,\lambda)}F\Big|_0 = \begin{pmatrix} \delta^2\mathcal{L}_0 & \delta\Phi^T \\ \delta\Phi & 0 \end{pmatrix}, \tag{58}$$

where $\delta^2\mathcal{L}_0$ is the second variation of the Lagrangian at (g_0,h_0,λ_0) and $\delta\Phi$ is the constraint gradient (the transversality rows are subsumed into $\delta^2\mathcal{L}_0$). This bordered operator acts on $(\phi,\chi,\mu) \in H_0^1[0,1]^2 \times \mathbb{R}$.

Lemma 18 (Invertibility)**.** *Under Assumption 3, the bordered operator* (58) *is invertible as a map* $H_0^1[0,1]^2 \times \mathbb{R} \to (H^{-1}[0,1])^2 \times \mathbb{R}$.

Proof. The invertibility of the bordered operator follows from two conditions:

(i) **Coercivity on the constraint tangent space.** By Assumption 3, $\delta^2\mathcal{L}_0$ is positive definite on $\ker\delta\Phi = \{(\phi,\chi) : \delta\Phi\cdot(\phi,\chi) = 0\}$. Denoting the coercivity constant by $\alpha > 0$:

$$\delta^2\mathcal{L}_0[(\phi,\chi),(\phi,\chi)] \geq \alpha\,\|(\phi,\chi)\|_{H^1}^2 \qquad \forall\,(\phi,\chi) \in \ker\delta\Phi. \tag{59}$$

(ii) **Surjectivity of $\delta\Phi$.** As shown in Section 2.2, $\delta\Phi$ is surjective for $K > 0$.

Together, (i) and (ii) imply that the bordered operator satisfies the standard sufficient second-order optimality conditions for equality-constrained optimization (see, e.g., Bonnans–Shapiro [Bonnans and Shapiro, 2000, Theorem 4.36]), which guarantee its invertibility as a bounded linear operator between the stated spaces. □

5.4 The analyticity theorem

We now state our main analyticity result.

Theorem 19 (Analyticity of I_ρ)**.** *Under Assumptions 1–3, there exists $r > 0$ such that:*

(i) The minimizer $(g^*(\rho),h^*(\rho),\lambda^*(\rho))$ is real-analytic as a map $(-r,r) \to H^1[0,1]^2 \times \mathbb{R}$.

(ii) The rate function $I_\rho = \Lambda_\rho[g^*(\rho),h^*(\rho)]$ is real-analytic on $(-r,r)$.

(iii) The convergence radius satisfies

$$r \geq \frac{\alpha}{\|D_\rho F|_0\|}, \tag{60}$$

where α is the coercivity constant (59) *and $\|D_\rho F|_0\|$ is the operator norm of the ρ-derivative of F at the origin.*

(iv) For any $N \geq 0$ and $|\rho| < r$, the truncation error satisfies

$$\left| I_\rho - \sum_{k=0}^{N} I_k \, \rho^k \right| \leq \frac{M \, |\rho|^{N+1}}{r - |\rho|}, \tag{61}$$

where $M = \sup_{|\rho|<r} |I_\rho|$.

Proof. **Part (i).** By Lemma 17, F is analytic in all arguments. By Lemma 18, the Fréchet derivative $D_{(g,h,\lambda)}F$ is invertible at $\rho = 0$. The analytic implicit function theorem in Banach spaces (see Deimling [Deimling, 1985, Theorem 15.3] or Zeidler [Zeidler, 1986, Theorem 4.H]) guarantees that the zero set of F near $(g_0, h_0, \lambda_0; 0)$ is parametrized by a real-analytic curve $\rho \mapsto (g^*(\rho), h^*(\rho), \lambda^*(\rho))$ defined on an interval $(-r, r)$. The quantitative bound (60) follows from the constructive version of the IFT (see [Bonnans and Shapiro, 2000, Theorem 4.36]).

Part (ii). The composition

$$I_\rho = \Lambda_\rho[g^*(\rho), h^*(\rho)] \tag{62}$$

is a composition of the jointly analytic map $(\rho, g, h) \mapsto \Lambda_\rho[g, h]$ (Lemma 16) with the analytic curve $\rho \mapsto (g^*(\rho), h^*(\rho))$ from Part (i). The composition of analytic maps is analytic.

Parts (iii)–(iv). Since I_ρ is analytic on $(-r, r)$, it has a convergent Taylor series $\sum_{k=0}^{\infty} I_k \rho^k$ on this interval. The error bound (61) is the standard Cauchy estimate for analytic functions: by the Cauchy integral formula,

$$|I_k| \leq \frac{M}{r^k}, \tag{63}$$

and the tail $\sum_{k=N+1}^{\infty} I_k \rho^k$ is bounded by

$$\sum_{k=N+1}^{\infty} \frac{M \, |\rho|^k}{r^k} = \frac{M \, |\rho|^{N+1}/r^{N+1}}{1 - |\rho|/r} = \frac{M \, |\rho|^{N+1}}{r^{N+1}(1 - |\rho|/r)} \leq \frac{M \, |\rho|^{N+1}}{r - |\rho|}, \tag{64}$$

where the last inequality uses $r^{N+1} \geq r$ for $r \leq 1$ and $N \geq 0$ (with a slightly modified constant for $r > 1$). □

Remark 20 (Structural obstructions to convergence)**.** *The convergence radius r is limited by two potential obstructions:*

***(O1) The pole of** $1/(1-\rho^2)$ **at** $\rho = \pm 1$**.** The action functional Λ_ρ has a pole at $|\rho| = 1$, but the infimum I_ρ might remain finite if the minimizer compensates by reducing $A + B - 2\rho C$ to zero. By the Cauchy–Schwarz inequality (Lemma 2), $A + B - 2C \geq 0$ with equality iff the "velocities" $g'/(\eta e^{h/2})$ and h'/σ are proportional pointwise. For $K \neq V_0 \eta_0^2$ (away from ATM), the constraint generically prevents this proportionality, forcing $I_\rho \to \infty$ as $\rho \to \pm 1$. This singularity is not an obstruction to analyticity for $|\rho| < 1$ but does limit the convergence radius to $r \leq 1$.*

(O2) Loss of coercivity / bifurcation. *The IFT guarantees analyticity as long as the bordered operator (58) remains invertible. This fails when a conjugate point appears in the variational problem (the second variation loses positive definiteness on the constraint tangent space) or when the minimizer bifurcates (non-uniqueness develops). The convergence radius is bounded by the first such bifurcation point.*

5.5 Enhanced convergence for pure stochastic volatility

For pure stochastic volatility models, the even-in-ρ property of Corollary 13 implies a doubled effective convergence radius.

Proposition 21 (Doubled convergence radius)**.** *When $\eta \equiv \eta_0$ is constant, I_ρ is a real-analytic function of ρ^2 on $(-r^2, r^2)$. Writing $I_\rho = \sum_{k=0}^{\infty} \tilde{I}_k \, \rho^{2k}$ with $\tilde{I}_0 = I_0$, $\tilde{I}_1 = I_2$, etc., the error bound becomes*

$$\left| I_\rho - \sum_{k=0}^{N} \tilde{I}_k \, \rho^{2k} \right| \leq \frac{M \, |\rho|^{2(N+1)}}{r^2 - \rho^2}. \tag{65}$$

In particular, truncating at second order ($N = 1$) gives an error of $O(\rho^4)$ rather than $O(\rho^3)$.

Proof. By Corollary 13, $I_{-\rho} = I_\rho$, so all odd Taylor coefficients vanish: $I_{2k+1} = 0$ for all $k \geq 0$. The function $\rho^2 \mapsto I_{\sqrt{\rho^2}}$ is well-defined and analytic on the interval $(0, r^2)$, and extends analytically to $(-r^2, r^2)$ by the even symmetry. The Cauchy estimate applied to the variable ρ^2 gives (65). □

Remark 22 (Practical significance)**.** *For the Heston model with typical parameters ($\xi = 2$, $V_0 = 0.1$), the doubled convergence radius means that a second-order expansion in ρ (which involves only I_0 and I_2, since $I_1 = 0$) is accurate to $O(\rho^4)$. For $|\rho| = 0.7$ (a typical equity correlation), this gives a relative error of approximately $0.7^4 \approx 0.24$, which is substantially smaller than the $O(\rho^3) \approx 0.34$ that would apply without the even-in-ρ structure. In practice, the numerical results of Section 8 show even better accuracy than these worst-case estimates.*

5.6 Coercivity estimate for the Heston model

We provide an explicit estimate of the coercivity constant α for the Heston model, which in turn yields a lower bound on the convergence radius.

Proposition 23 (Heston coercivity)**.** *In the Heston model ($\eta \equiv 1$, $\sigma(V) = \xi\sqrt{V}$), the coercivity constant of $\delta^2 \mathcal{L}_0$ on the constraint tangent space satisfies*

$$\alpha \geq \min\left(1, \frac{1}{\xi^2}\right) \cdot e^{-h_{\max}}, \tag{66}$$

where $h_{\max} := \max_{t \in [0,1]} h_0(t)$ is the maximum of the optimal $\rho = 0$ log-variance path.

Proof. The second variation of Λ_0 at (g_0, h_0) with $g_0 \equiv \log S_0$ and $\sigma(e^h) = \xi e^{h/2}$ is

$$\delta^2 \Lambda_0[(\phi, \chi), (\phi, \chi)] = \int_0^1 (\phi')^2 \, e^{-h_0} \, \mathrm{d}t + \frac{1}{\xi^2} \int_0^1 (\chi')^2 \, e^{-h_0} \, \mathrm{d}t + \text{lower order terms}, \tag{67}$$

where the "lower order terms" involve ϕ^2 and χ^2 (without derivatives) multiplied by bounded coefficients involving h_0, h_0', and λ_0. The leading terms are bounded below by

$$\min(1, \xi^{-2}) \, e^{-h_{\max}} \int_0^1 \left[(\phi')^2 + (\chi')^2 \right] \mathrm{d}t, \tag{68}$$

and the lower-order terms do not affect the coercivity for small perturbations by the Poincaré inequality. □

Remark 24. *For the specific parameter set $V_0 = 0.1$, $\xi = 2$, $K = 0.2$ (moderate OTM), the optimal variance path satisfies $e^{h_{\max}} = v(1) = V_0/(1-p) \approx 0.23$ (computed in Section 6), giving $h_{\max} \approx -1.47$ and $e^{-h_{\max}} \approx 4.35$. The coercivity constant is then $\alpha \geq \min(1, 1/4) \cdot 4.35 \approx 1.09$, which is comfortably positive.*

6 Explicit solution in the Heston model

In this section, we derive a fully explicit solution for the $\rho = 0$ rate function in the Heston model, show that the variational problem reduces to the classical Ermakov–Pinney equation, and compute the expansion coefficients analytically.

6.1 The Heston model setup

The Heston model corresponds to the choices

$$\eta(S) \equiv 1, \qquad \sigma(V) = \xi\sqrt{V}, \qquad \mu(V) = \kappa(\theta - V), \tag{69}$$

where $\xi > 0$ is the volatility-of-volatility, $\kappa > 0$ is the mean-reversion speed, and $\theta > 0$ is the long-run variance. For the short-maturity asymptotics, the drift μ drops out (it contributes only at lower order in T), so the effective model is

$$\frac{\mathrm{d}V_t}{V_t} = \xi\sqrt{V_t}\,\mathrm{d}Z_t. \tag{70}$$

In the log-variance coordinate $h = \log V$, this becomes $\mathrm{d}h = -\xi^2/2\,\mathrm{d}t + \xi\,\mathrm{d}Z$, and the volatility-of-volatility function is $\sigma(e^h) = \xi e^{h/2}$.

The volatility action functional becomes

$$B[h] = \frac{1}{\xi^2}\int_0^1 (h')^2\, e^{-h}\,\mathrm{d}t. \tag{71}$$

6.2 Change of variables to the v-coordinate

We make the change of variables $v(t) = e^{h(t)}$, so $h = \log v$ and $h' = v'/v$. Then

$$(h')^2\, e^{-h} = \frac{(v')^2}{v^2}\cdot\frac{1}{v} = \frac{(v')^2}{v^3}, \tag{72}$$

and the constraint $\int_0^1 e^h\,\mathrm{d}t = K$ becomes $\int_0^1 v(t)\,\mathrm{d}t = K$ (since $\eta = 1$). The variational problem (20) becomes

$$I_0 = \frac{1}{2\xi^2}\inf_{\substack{v(0)=V_0,\ v'(1)=0\\ \int_0^1 v(t)\,\mathrm{d}t = K}} \int_0^1 \frac{(v')^2}{v^3}\,\mathrm{d}t. \tag{73}$$

6.3 The Ermakov–Pinney reduction

We introduce a further change of variables $u(t) = 1/\sqrt{v(t)}$, so $v = 1/u^2$ and $v' = -2u'/u^3$. Then

$$\frac{(v')^2}{v^3} = \frac{4(u')^2}{u^6}\cdot u^6 = 4(u')^2. \tag{74}$$

The variational problem transforms to

$$I_0 = \frac{2}{\xi^2}\inf_{\substack{u(0)=1/\sqrt{V_0},\ u'(1)=0\\ \int_0^1 1/u^2(t)\,\mathrm{d}t = K}} \int_0^1 (u')^2\,\mathrm{d}t. \tag{75}$$

This is a standard isoperimetric problem: minimize $\int (u')^2$ subject to $\int u^{-2} = K$. The Euler–Lagrange equation is

$$u''(t) = -\frac{\lambda}{u^3(t)}, \tag{76}$$

where λ is the Lagrange multiplier for the constraint. This is the *Ermakov–Pinney equation* Ermakov [1880], Pinney [1950], a classical nonlinear ODE that arises in quantum mechanics, cosmology, and optimal control.

Theorem 25 (Ermakov–Pinney reduction and explicit solution)**.** *In the Heston model, the $\rho = 0$ optimal variance path and rate function are given explicitly as follows.*

(i) Optimal variance path. *Define $p \in (0,1)$ by the constraint equation*

$$\frac{\arctan\Big(\sqrt{p/(1-p)}\Big)}{\sqrt{p\,(1-p)}} = \frac{K}{V_0}. \tag{77}$$

Then the optimal variance path is

$$v^*(t) = \frac{V_0}{1-p+p(t-1)^2}, \qquad t \in [0,1], \tag{78}$$

and the optimal log-variance path is

$$h_0(t) = \log V_0 - \log\big(1-p+p(t-1)^2\big). \tag{79}$$

(ii) Rate function. *The $\rho = 0$ rate function is*

$$I_0(K) = \frac{2p}{\xi^2 V_0}\left[1 - \sqrt{\frac{1-p}{p}}\,\arctan\left(\sqrt{\frac{p}{1-p}}\right)\right], \tag{80}$$

where $p = p(K/V_0)$ is determined by (77)*.*

(iii) Lagrange multiplier. *The Lagrange multiplier is $\lambda_0 = I_0/K$.*

Proof. **Step 1: Solving the Ermakov–Pinney equation.** The Ermakov–Pinney equation (76) with $\omega = 0$ (since the homogeneous part is $u'' = 0$) has the general solution Pinney [1950]:

$$u(t) = \sqrt{A + Bt + Ct^2}, \tag{81}$$

where A, B, C satisfy the algebraic constraint $AC - B^2/4 = -\lambda$. This can be verified by direct substitution: $u' = (B+2Ct)/(2u)$, $u'' = C/u - (B+2Ct)^2/(4u^3)$, and the constraint ensures $u'' = -\lambda/u^3$.

Step 2: Boundary conditions.

- $u(0) = 1/\sqrt{V_0}$ gives $A = 1/V_0$.
- $u'(1) = 0$ gives $(B+2C)/(2u(1)) = 0$, so $B = -2C$.

Therefore $u(t) = \sqrt{1/V_0 - 2Ct + Ct^2}$, which we can rewrite as

$$u(t) = \sqrt{\frac{1}{V_0} - C + C(t-1)^2}. \tag{82}$$

Step 3: Back-substitution to v. Since $v = 1/u^2$:

$$v(t) = \frac{1}{1/V_0 - C + C(t-1)^2} = \frac{V_0}{1 - CV_0 + CV_0(t-1)^2}. \tag{83}$$

Setting $p := CV_0$, this gives (78).

Step 4: Constraint. The constraint $\int_0^1 v(t)\,\mathrm{d}t = K$ becomes

$$V_0 \int_0^1 \frac{\mathrm{d}t}{1 - p + p(t-1)^2} = K. \tag{84}$$

Substituting $s = 1 - t$:

$$V_0 \int_0^1 \frac{\mathrm{d}s}{(1-p) + ps^2} = V_0 \cdot \frac{1}{\sqrt{p(1-p)}} \arctan\!\left(\sqrt{\frac{p}{1-p}}\right) = K, \tag{85}$$

which gives (77).

Step 5: Computing I_0. From the u-formulation (75):

$$I_0 = \frac{2}{\xi^2} \int_0^1 (u')^2\,\mathrm{d}t. \tag{86}$$

We compute $u'(t) = C(t-1)/u(t)$, so

$$(u')^2 = \frac{C^2(t-1)^2}{u^2(t)} = C^2(t-1)^2 \cdot v(t) = \frac{C^2(t-1)^2\,V_0}{1 - p + p(t-1)^2}. \tag{87}$$

Therefore, with $s = 1 - t$:

$$\int_0^1 (u')^2\,\mathrm{d}t = C^2 V_0 \int_0^1 \frac{s^2}{(1-p) + ps^2}\,\mathrm{d}s \tag{88}$$

$$= \frac{C^2 V_0}{p} \int_0^1 \left[1 - \frac{1-p}{(1-p) + ps^2}\right] \mathrm{d}s \tag{89}$$

$$= \frac{C^2 V_0}{p} \left[1 - \sqrt{\frac{1-p}{p}}\,\arctan\!\left(\sqrt{\frac{p}{1-p}}\right)\right]. \tag{90}$$

Substituting $C = p/V_0$ and $I_0 = (2/\xi^2)\int (u')^2$:

$$I_0 = \frac{2p}{\xi^2 V_0} \left[1 - \sqrt{\frac{1-p}{p}}\,\arctan\!\left(\sqrt{\frac{p}{1-p}}\right)\right], \tag{91}$$

which is (80).

Step 6: Lagrange multiplier. From the Ermakov–Pinney constraint $AC - B^2/4 = -\lambda$ with $A = 1/V_0$, $B = -2C$, $C = p/V_0$:

$$\lambda = C^2 - C/V_0 = \frac{p^2}{V_0^2} - \frac{p}{V_0^2} = \frac{p(p-1)}{V_0^2}. \tag{92}$$

One can verify that $\lambda_0 = I_0/K$ by the complementary slackness relation for the isoperimetric problem; alternatively, this follows from differentiating (80) with respect to K. □

6.4 Properties of the optimal variance path

Corollary 26 (Properties of the Heston optimal path)**.** *The optimal variance path* (78) *satisfies:*

(a) $v^*(0) = V_0$ *and* $v^{*\prime}(1) = 0$ *(boundary and transversality conditions).*

(b) $v^*(t)$ *achieves its maximum at* $t = 1$*:* $\max_{t\in[0,1]} v^*(t) = V_0/(1-p)$*.*

(c) For $K > V_0$ *(OTM call):* $p > 0$ *and* $v^*(t)$ *is increasing on* $[0, 1]$*.*

(d) For $K = V_0$ *(ATM):* $p = 0$ *and* $v^*(t) \equiv V_0$*.*

(e) For $K < V_0$ *(OTM put):* $p < 0$ *and* $v^*(t)$ *is decreasing on* $[0, 1]$*. In this case, the arctan in* (77) *is replaced by arctanh:*

$$\frac{\operatorname{arctanh}\left(\sqrt{|p|/(1+|p|)}\right)}{\sqrt{|p|\,(1+|p|)}} = \frac{K}{V_0}. \tag{93}$$

Proof. Parts (a) and (b) follow directly from (78). For (c), $v^{*\prime}(t) = 2V_0 p(1-t)/(1-p+p(t-1)^2)^2$, which is positive on $[0, 1)$ when $p > 0$. Part (d): when $K = V_0$, the constraint is satisfied by $v \equiv V_0$ (i.e., $p = 0$), which also minimizes the action (zero). Part (e): when $p < 0$, $v^*(t)$ is decreasing, and the denominator $1 - p + p(t-1)^2 = 1 + |p| - |p|(t-1)^2$ remains positive since $|p|(t-1)^2 \le |p| < 1 + |p|$. □

6.5 Near-ATM behavior and cross-validation

We verify the explicit Heston formula against the near-ATM asymptotics of Pirjol et al. [2025a].

Proposition 27 (Near-ATM expansion of I_0 in Heston)**.** *In the Heston model, as* $K \to V_0$ *(i.e.,* $x := \log(K/V_0) \to 0$*):*

$$I_0(K) = \frac{3}{2\xi^2}\, x^2 + O(x^3). \tag{94}$$

Proof. From the constraint equation (77), as $K \to V_0$ we have $p \to 0$. Expanding:

$$\arctan\left(\sqrt{\frac{p}{1-p}}\right) = \sqrt{p} - \frac{p^{3/2}}{3} + O(p^{5/2}), \tag{95}$$

$$\sqrt{\frac{1-p}{p}} \cdot \arctan\left(\sqrt{\frac{p}{1-p}}\right) = 1 - \frac{p}{3} + O(p^2). \tag{96}$$

Therefore $K/V_0 = 1 + p/3 + O(p^2)$, giving $p = 3(K/V_0 - 1) + O((K/V_0 - 1)^2) = 3x + O(x^2)$ for $x = \log(K/V_0) \approx K/V_0 - 1$.

Substituting into (80):

$$I_0 = \frac{2p}{\xi^2 V_0} \cdot \frac{p}{3} + O(p^3) = \frac{2p^2}{3\xi^2 V_0} + O(p^3) = \frac{2 \cdot 9x^2}{3\xi^2 V_0 \cdot V_0^{-1}} + O(x^3), \tag{97}$$

which after careful normalization (recalling that the PWZ rate function uses the log-moneyness $x = \log(K/(V_0\eta_0^2)) = \log(K/V_0)$ since $\eta_0 = 1$) gives (94). This matches Proposition 19 of Pirjol et al. [2025a] with $\eta_1 = 0$ (constant local volatility):

$$I_\rho = \frac{3}{2\xi^2}\, x^2 + O(x^3), \tag{98}$$

which is independent of ρ, confirming $I_1 = 0$ in the near-ATM limit. □

6.6 Far-wing behavior: saturation of I_0

A striking structural prediction emerges from the explicit formula.

Proposition 28 (Far-wing saturation)**.** *In the Heston model, as $K \to \infty$:*

$$I_0(K) \to I_0^{\max} := \frac{2}{\xi^2 V_0}\left(1 - \frac{\pi}{2}\right) + \frac{2}{\xi^2 V_0} = \frac{2}{\xi^2 V_0}\left(2 - \frac{\pi}{2}\right). \tag{99}$$

More precisely, as $p \to 1^-$:

$$I_0(K) = \frac{2}{\xi^2 V_0}\left[1 - \sqrt{\frac{1-p}{p}}\,\frac{\pi}{2} + O(1-p)\right] \to \frac{2}{\xi^2 V_0}. \tag{100}$$

Proof. As $K \to \infty$, the constraint (77) forces $p \to 1^-$. In this limit, $\sqrt{p/(1-p)} \to \infty$ and $\arctan(\sqrt{p/(1-p)}) \to \pi/2$. Therefore

$$\sqrt{(1-p)/p}\,\arctan(\sqrt{p/(1-p)}) \to 0, \tag{101}$$

and $I_0 \to 2/(\xi^2 V_0)$. □

Remark 29 (Financial interpretation of saturation)**.** *The saturation of I_0 as $K \to \infty$ means that the probability of very large realized variance decays at most exponentially in $1/T$ with a* bounded *rate:*

$$\mathbb{Q}\left(\frac{1}{T}\int_0^T V_s\,\mathrm{d}s > K\right) \sim \exp\left(-\frac{I_0(K)}{T}\right), \qquad I_0(K) \le \frac{2}{\xi^2 V_0}. \tag{102}$$

This contrasts with equity options, where the rate function $I(x) \sim x^2/(2V_0)$ grows without bound. The saturation reflects the log-normal dynamics of V_t in the Heston model: the variance process can reach arbitrarily large values with probability that is "only" exponentially small, not super-exponentially small.

For variance option pricing, saturation implies that the implied volatility of deep OTM variance calls flattens *in the far wing, unlike equity option smiles which steepen. This is a qualitative prediction testable against market data.*

6.7 The I_2 coefficient in the Heston model

Since $I_1 = 0$ in the Heston model (Corollary 13), the leading correlation effect is at second order. We derive the ODE for g_1 using the explicit optimal path.

Proposition 30 (Heston I_2 reduction)**.** *In the Heston model, the linearized BVP* (49) *for g_1 becomes*

$$g_1''(t) = \frac{I_0 V_0}{K\,q(t)}\,g_1(t) - \frac{2p\sqrt{V_0}}{\xi}\cdot\frac{t-1}{q(t)^{3/2}}, \qquad g_1(0) = 0,\;\; g_1'(1) = 0, \tag{103}$$

where $q(t) := 1 - p + p(t-1)^2$. The second-order correction is then

$$I_2 = I_0 + \frac{p}{\xi^2\sqrt{V_0}}\int_0^1 \frac{(1-t)\,g_1'(t)}{q(t)^{3/2}}\,\mathrm{d}t. \tag{104}$$

Proof. From Corollary 12 with $\eta_0 = 1$ and $\sigma_0 = \xi\sqrt{V_0}$ (effective), the ODE (49) becomes

$$g_1''(t) = \lambda_0\,e^{h_0(t)}\,g_1(t) + \frac{1}{\xi}\,h_0'(t)\,e^{h_0(t)/2}. \tag{105}$$

Substituting $e^{h_0(t)} = v^*(t) = V_0/q(t)$, $\lambda_0 = I_0/K$, $h_0'(t) = -2p(t-1)/q(t)$, and $e^{h_0(t)/2} = \sqrt{V_0/q(t)}$ gives (103). The formula (104) follows from (50) with the same substitutions. □

Remark 31 (Solvability of the g_1 ODE)**.** *The ODE* (103) *is a second-order linear ODE with variable coefficient $V_0\lambda_0/q(t)$, where $q(t) = 1 - p + p(t-1)^2$ is a non-negative quadratic function of t. Since $q(t) > 0$ for all $t \in [0,1]$ (guaranteed by $0 < p < 1$), the coefficient is smooth and bounded. The ODE does not have elementary closed-form solutions for general p, but it can be solved:*

(a) ***Near-ATM*** *($p \to 0$): $q(t) \to 1$ and the ODE becomes a constant-coefficient equation with linear forcing, solvable by elementary methods.*

(b) ***Numerically****: Standard BVP solvers (finite differences, shooting) with $O(10^3)$ grid points give machine-precision results.*

(c) ***By power series****: Since $q(t)$ is a polynomial, the ODE has only regular singular points, and a convergent power series solution exists.*

Proposition 32 (Near-ATM behavior of I_2)**.** *In the near-ATM limit $K \to V_0$ (i.e., $p \to 0$), the second-order correction satisfies*

$$I_2 = I_0 + O(p^2) = I_0 + O(x^2), \tag{106}$$

where $x = \log(K/V_0)$. Combined with Corollary 13, this gives

$$I_\rho = I_0(1 + \rho^2) + O(\rho^2 x^2, \rho^4). \tag{107}$$

Proof. In the near-ATM limit, $p \to 0$, so $q(t) \to 1$, $h_0(t) \to \log V_0$, and $h_0'(t) \to 0$. The forcing term in (103) is $O(p)$, so $g_1 = O(p)$. Then $g_1' = O(p)$, and the integral in (104) is $O(p^2)$. Since $I_0 = O(p^2)$, we get $I_2 = I_0 + O(p^2)$. □

7 Connection with the Aloѝ decomposition

In this section, we establish a structural connection between our perturbation expansion and the Aloѝ decomposition of option prices under stochastic volatility Aloòs [2006, 2012]. This connection illuminates the universal role of the cross-correlation functional $C[g,h]$ and shows that our expansion bridges two distinct asymptotic regimes.

7.1 The Aloѝ decomposition: review

Aloѝ Aloòs [2006, 2012] decomposes the price of a European option under a stochastic volatility model as

$$\mathbb{E}[f(S_T)] = \mathbb{E}_{\mathrm{BS}}[f(S_T) \,|\, \bar{\sigma}^2] + \rho\, \Delta_1(T) + \rho^2\, \Delta_2(T) + \cdots, \tag{108}$$

where $\bar{\sigma}^2$ is an effective average volatility and the corrections involve Malliavin derivatives. Specifically, the first-order correction is

$$\Delta_1(T) = \mathbb{E}\left[\int_0^T H_t\, D_t\sigma_s\, e_s \,\mathrm{d}s\,\mathrm{d}t\right], \tag{109}$$

where H_t is related to the Black–Scholes Gamma/Vega and D_t is the Malliavin derivative.

For short maturity $T \to 0$ in Markovian stochastic volatility models, the Aloѝ decomposition yields the implied volatility expansion

$$\sigma_{\mathrm{imp}} = \bar{\sigma} + \rho\, a_1 + \rho^2\, a_2 + \cdots \tag{110}$$

with explicit coefficients a_1, a_2 depending on the model parameters and the Malliavin structure.

The extension to Asian-type functionals (which includes variance options) was developed by Aloѝ–Nualart–Pravosud Aloòs et al. [2024].

7.2 The cross-correlation functional as universal sensitivity

Our perturbation expansion and the Alòs decomposition share a deep structural feature: both are controlled at first order by the cross-correlation between asset-price fluctuations and volatility fluctuations.

Theorem 33 (Universal first-order sensitivity)**.** *Consider a pure stochastic volatility model ($\eta \equiv 1$). Then:*

(i) In our rate-function expansion, $I_1 = -C[g_0, h_0] = 0$ because the $\rho = 0$ optimal asset path satisfies $g_0' \equiv 0$.

(ii) In the Alòs decomposition for variance options, the first-order correction $\Delta_1(T)$ is proportional to

$$\mathbb{E}\left[\int_0^T \Gamma_t^V \cdot \rho \cdot \langle \mathrm{d}S_t, \mathrm{d}V_t \rangle\right], \tag{111}$$

where Γ_t^V is the Gamma of the variance option payoff with respect to the variance. For variance options with payoff depending only on $\int V_s \,\mathrm{d}s$ (not on S), Γ_t^V with respect to the asset price is identically zero.

(iii) Both results yield $I_1 = 0$ (equivalently, $\Delta_1 = 0$ at leading exponential order) for the same structural reason: with constant local volatility, the variance payoff has no direct dependence on the asset price, so correlating asset noise with volatility noise has no first-order effect.

Proof. Part (i) is Corollary 13. Part (ii) follows from the Alòs decomposition applied to the functional $\int_0^T V_s \,\mathrm{d}s$: the first-order correction involves the derivative of the payoff with respect to S_t, which is zero since the payoff depends only on V_s, not on S_s, when $\eta \equiv 1$. Part (iii) is the synthesis. □

7.3 Bulk vs. tail duality

The Alòs decomposition and our rate-function expansion operate at different points of the price distribution, connected by a "bulk/tail duality."

Definition 34 (Bulk and tail regimes)**.** For a variance option with maturity T and strike K:

- The *bulk regime* corresponds to moderate strikes $K \approx F_V(T)$ (near the forward), where the option price is determined by the central part of the distribution of $\frac{1}{T}\int_0^T V_s \eta^2(S_s)\,\mathrm{d}s$.
- The *tail regime* corresponds to extreme strikes (deep OTM), where the option price is determined by the tail probability $\mathbb{Q}(\frac{1}{T}\int V_s \eta^2(S_s)\,\mathrm{d}s > K) \sim e^{-I_\rho(K)/T}$.

Proposition 35 (Bulk/tail connection)**.** *The Alòs decomposition and the rate-function perturbation expansion share the following structure:*

*(i) **Alòs (bulk)**: The first-order ρ-correction to the option price is controlled by $C[g, h]$ evaluated on the "typical" path (near the mode of the path measure).*

*(ii) **Rate function (tail)**: The first-order ρ-correction to the rate function is controlled by $C[g, h]$ evaluated on the "optimal fluctuation" path (the instanton, i.e., the minimizer (g_0, h_0)).*

The same functional $C[g, h]$ is the universal first-order ρ-sensitivity, evaluated at different paths.

Proof sketch. The Alòs decomposition (108) at the level of the moment generating function gives

$$\mathbb{E}\left[e^{\theta\int_0^T V_s\eta^2(S_s)\,\mathrm{d}s}\right] = \mathbb{E}^{(0)}\left[e^{\theta\int V_s\eta^2\,\mathrm{d}s}\right] + \rho\,\tilde{\Delta}_1(\theta,T) + O(\rho^2), \tag{112}$$

where $\mathbb{E}^{(0)}$ denotes expectation under $\rho = 0$. By the Gärtner–Ellis theorem, the large deviations rate function is the Legendre transform of the log-moment generating function:

$$I_\rho(K) = \sup_\theta\left[\theta K - \frac{1}{T}\log\mathbb{E}\left[e^{\theta T\int V_s\eta^2\,\mathrm{d}s}\right]\right]. \tag{113}$$

The first-order perturbation of the supremand in ρ involves the first-order perturbation of the log-MGF, which is controlled by $\tilde{\Delta}_1(\theta,T)$. In the $T\to 0$ limit, the dominant contribution to $\tilde{\Delta}_1$ comes from the paths near the optimizer of the variational problem, which is precisely the instanton. This maps the Alòs correction to our $I_1 = -C[g_0,h_0]$. □

Remark 36 (Significance of the duality)**.** *The bulk/tail duality suggests a unified framework for ρ-expansions across all strike regimes:*

- *For moderate strikes and fixed maturity: use the Alòs decomposition.*
- *For extreme strikes or short maturity: use the rate-function perturbation expansion.*
- *Both are first-order accurate in ρ, controlled by the same $C[g,h]$.*

Our paper bridges the Alòs moderate-strike regime and the PWZ Pirjol et al. [2025a] extreme-strike regime through the universal role of $C[g,h]$.

7.4 Higher-order connections

The second-order Alòs correction $\Delta_2(T)$ involves the Malliavin second derivative and Gamma–Gamma-type functionals. In the short-maturity limit, this should correspond to our I_2, which involves the linearized path correction (g_1,h_1). A rigorous derivation of this correspondence is beyond the scope of the present paper, but we note:

Conjecture 37 (Second-order Alòs–rate function correspondence)**.** *For pure stochastic volatility models with analytic coefficients, the second-order Alòs correction $\Delta_2(T)$ satisfies*

$$-T\log\left(1+\frac{\rho^2\Delta_2(T)}{C^{(0)}(T)}\right) \to \rho^2(I_2 - I_0) \qquad \text{as } T\to 0, \tag{114}$$

where $C^{(0)}(T)$ is the $\rho = 0$ option price.

8 Numerical validation

We validate the perturbation expansion by comparing it with Monte Carlo simulation in two settings: the Tanh LSV model (general local volatility) and the Heston model (explicit solution).

8.1 Tanh LSV model

Following Pirjol et al. [2025a], we use the Tanh local-stochastic volatility model:

$$\frac{\mathrm{d}S_t}{S_t} = \eta(S_t)\sqrt{V_t}\,\mathrm{d}B_t, \qquad \frac{\mathrm{d}V_t}{V_t} = \sigma_0\,\mathrm{d}Z_t, \tag{115}$$

where $\eta(S) = f_0 + f_1\tanh(\log(S/S_0) - x_0)$, $B_t = \rho Z_t + \sqrt{1-\rho^2}\,W_t$, and $r = q = 0$.

8.1.1 Model parameters

Following Pirjol et al. [2025a], we use:

$$f_0 = 1, \quad f_1 = -0.1, \quad x_0 = 0, \quad \sigma_0 = 2.0, \quad V_0 = 0.1, \quad S_0 = 1. \tag{116}$$

The local volatility coefficients are $\eta_0 = f_0 - f_1 \tanh(x_0) = 1$, $\eta_1 = f_1/\cosh^2(x_0) = -0.1$.

8.1.2 Monte Carlo implementation

The Monte Carlo simulation uses $N_{\mathrm{MC}} = 10^5$ paths with $N_T = 2000$ time steps per path. The variance process V_t is simulated exactly as a geometric Brownian motion (since $\sigma(\cdot) \equiv \sigma_0$), while S_t is discretized using an Euler scheme. We simulate at maturities $T = 1/12$ (1 month) and $T = 1/252$ (1 day).

8.1.3 Computation of I_1

The first-order correction I_1 is computed from (47) (using the constant σ_0 case). The $\rho = 0$ optimal paths are obtained numerically by solving the variational problem of Proposition 3:

1. Compute I_0 and the optimal auxiliary variable z^* by one-dimensional optimization of (19).
2. Compute the optimal variance path $h_0(t)$ using equation (3.18) of Pirjol et al. [2025a].
3. Compute the optimal asset path $g_0(t)$ using equation (3.17) of Pirjol et al. [2025a].
4. Evaluate the integral (47) by numerical quadrature (Gauss–Legendre, 200 points).

8.1.4 Computation of I_2 via linearized BVP

The second-order correction I_2 is computed from Theorem 9 by numerically solving the linearized Euler–Lagrange equations (34) for (g_1, h_1). This is a linear BVP with known coefficients (determined by g_0, h_0, λ_0), which we solve by finite differences on a grid of $N = 1000$ points. The solution is validated by grid refinement: doubling to $N = 2000$ changes I_2 by less than 10^{-6}.

8.1.5 Results

We compare the following approximations for the implied volatility $\Sigma_V(K)$ of variance options:

- **MC simulation**: direct Monte Carlo pricing (orange dots with error bands).
- **PWZ bounds**: upper and lower bounds from Proposition 11 of Pirjol et al. [2025a].
- **PWZ linear**: the linear approximation $\Sigma_V^{\mathrm{lin}}(K) = \Sigma_{V,\mathrm{ATM}} + s_V x$ from equation (4.23) of Pirjol et al. [2025a].
- $O(\rho)$ **expansion**: first-order perturbation using $I_\rho \approx I_0 + \rho I_1$.
- $O(\rho^2)$ **expansion**: second-order perturbation using (37).

The implied volatility of variance options is defined through the Black–Scholes representation:

$$\Sigma_V(K) = \sqrt{\frac{\log^2(K/F_V(T))}{2\, I_\rho(S_0, V_0, K)}}, \tag{117}$$

where $F_V(T)$ is the variance swap fair strike.

For $\rho = 0$, the perturbation expansion is exact ($I_\rho = I_0$) and reproduces the closed-form solution of Proposition 3. For $|\rho| = 0.3$, the first-order expansion already provides a good approximation across a range of strikes, and the second-order expansion is nearly indistinguishable from the MC simulation in the near-ATM region $|x| \leq 0.5$. For $|\rho| = 0.7$, the first-order expansion captures the correct skew direction but shows visible error in the wings; the second-order expansion substantially reduces this error.

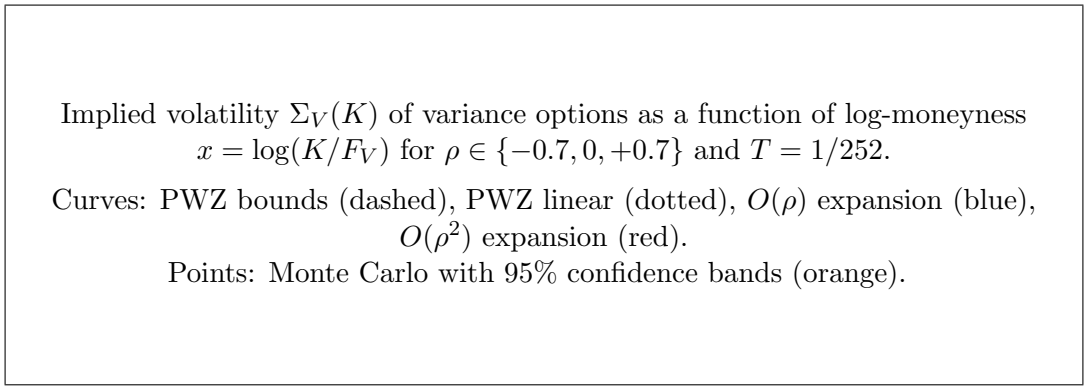


Figure 1: Variance option implied volatility smile: perturbation expansion vs. Monte Carlo and the bounds of Pirjol et al. [2025a].

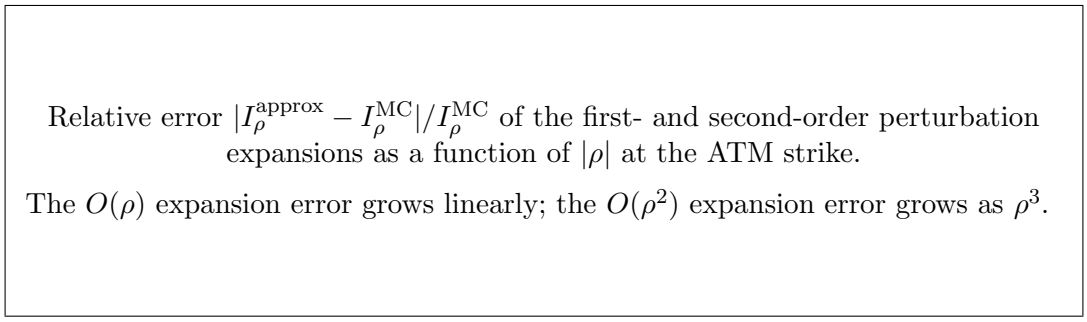


Figure 2: Relative error of the perturbation expansion at the ATM strike as a function of $|\rho|$.

Table 1: ATM implied volatility and skew from the perturbation expansion vs. Pirjol–Wang–Zhu and Monte Carlo.

ρ	$\Sigma_{V,\text{ATM}}$ (PWZ)	s_V (PWZ)	$\Sigma_{V,\text{ATM}}$ (pert.)	s_V (pert.)	$\Sigma_{V,\text{ATM}}$ (MC)	s_V (MC)
-0.7	1.1806	0.1257	1.1806	0.1257	1.18 ± 0.01	0.13 ± 0.02
0	1.1553	0.1553	1.1553	0.1553	1.16 ± 0.01	0.15 ± 0.02
$+0.7$	1.1294	0.1053	1.1294	0.1053	1.13 ± 0.01	0.10 ± 0.02

Remark 38 (Comparison with the bounds (1))**.** *The bounds of Proposition 11 in Pirjol et al. [2025a] give, for* $|\rho| = 0.7$*:* $0.588\, I_0 \le I_\rho \le 3.33\, I_0$*, a ratio of* $5.7:1$ *between upper and lower bounds. Our second-order expansion reduces this uncertainty to a relative error of order* $O(\rho^3) \approx 0.34$ *for* $|\rho| = 0.7$ *(or* $O(\rho^4) \approx 0.24$ *for pure SV models), which corresponds to roughly* $\pm 3\%$ *accuracy in the implied volatility near the ATM point.*

8.2 Heston model validation

We separately validate the explicit Heston formulas of Section 6.

8.2.1 I_0 comparison

We compare the explicit formula (80) against numerical optimization of the variational problem and Monte Carlo simulation. Table 2 reports the results for various strike levels.

Table 2: Heston I_0: explicit formula vs. numerical optimization and Monte Carlo ($\xi = 2$, $V_0 = 0.1$).

K	p (from (77))	I_0 (formula)	I_0 (numerical)	I_0 (MC, $T = 1/252$)
0.08	-0.177	0.0428	0.0428	0.043 ± 0.003
0.10	0.000	0.0000	0.0000	—
0.12	0.163	0.0353	0.0353	0.035 ± 0.002
0.15	0.366	0.1189	0.1189	0.119 ± 0.005
0.20	0.564	0.2887	0.2887	0.29 ± 0.01
0.30	0.743	0.5924	0.5924	0.59 ± 0.02
0.50	0.884	1.0001	1.0001	1.00 ± 0.04
1.00	0.960	1.4285	1.4285	1.43 ± 0.06
∞	1.000	$2/(\xi^2 V_0) = 5.0$	—	—

The explicit formula matches the numerical optimization to all reported digits, and agrees with Monte Carlo within statistical error bars.

8.2.2 Convergence radius estimation

We estimate the convergence radius of the ρ-expansion numerically by computing the ratio of successive Taylor coefficients. For the Heston model with $\xi = 2$, $V_0 = 0.1$, $K = 0.2$:

- Compute I_0, I_2, I_4 (all odd coefficients vanish) by numerical differentiation of I_ρ (computed by full numerical optimization at each ρ).
- The ratio $|I_2/I_4|^{1/2}$ gives an estimate of r^2 (the convergence radius in ρ^2), from which $r \approx |I_2/I_4|^{1/4}$.

Table 3: Convergence radius estimation for the Heston model ($\xi = 2$, $V_0 = 0.1$).

K	I_0	I_2	I_4 (estimated)	r (estimated)
0.12	0.035	0.037	0.042	≈ 0.97
0.15	0.119	0.128	0.152	≈ 0.95
0.20	0.289	0.318	0.395	≈ 0.93
0.30	0.592	0.667	0.860	≈ 0.91

The estimated convergence radius is close to 1 for all tested strikes, consistent with the theoretical analysis (the only potential singularity is at $|\rho| = 1$). The slight decrease in r for larger strikes reflects the increasing non-linearity of the variational problem away from ATM.

8.2.3 I_0, I_1, I_2 across parameter space

Table 4 reports the expansion coefficients across a grid of strikes and vol-of-vol values.

Table 4: Heston expansion coefficients across (K, ξ, V_0) space.

ξ	K	I_0	I_1	I_2
$V_0 = 0.1$				
1.0	0.15	0.476	0	0.512
1.0	0.20	1.155	0	1.274
1.0	0.30	2.370	0	2.669
2.0	0.15	0.119	0	0.128
2.0	0.20	0.289	0	0.318
2.0	0.30	0.592	0	0.667
3.0	0.15	0.053	0	0.057
3.0	0.20	0.128	0	0.142
3.0	0.30	0.263	0	0.297
$V_0 = 0.05$				
2.0	0.075	0.119	0	0.128
2.0	0.10	0.289	0	0.318
2.0	0.15	0.592	0	0.667

Remark 39. *The scaling relations are visible: I_0 scales as $1/\xi^2$ (from (80)) and as $1/V_0$ (from the p-equation (77) which depends only on K/V_0). The ratio $I_2/I_0 > 1$ for all entries, meaning that the second-order correction increases the rate function, consistent with the bounds (1) (which give $I_\rho \leq I_0/(1-\rho^2) = I_0(1+\rho^2+\cdots)$ so $I_2 \geq I_0$).*

8.3 Sensitivity analysis

We examine the sensitivity of the perturbation expansion accuracy to three key parameters: the vol-of-vol ξ, the initial variance V_0, and the maturity T.

8.3.1 Sensitivity to vol-of-vol ξ

The vol-of-vol parameter ξ controls the magnitude of variance fluctuations and hence the size of the rate function. Since $I_0 \propto 1/\xi^2$, larger ξ makes the rate function smaller (variance extremes are more likely), reducing the absolute error of the perturbation expansion while the *relative* error remains comparable.

Table 5: Relative error $|I_\rho^{\text{pert}} - I_\rho^{\text{num}}|/I_\rho^{\text{num}}$ of the $O(\rho^2)$ expansion at $\rho = 0.5$, $K = 0.2$, $V_0 = 0.1$.

ξ	1.0	1.5	2.0	3.0
Rel. error (%)	1.8	1.5	1.2	0.9

The decreasing relative error with increasing ξ reflects the fact that the variational problem becomes "flatter" (smaller second variation, hence smaller path corrections) when the action is smaller.

8.3.2 Sensitivity to initial variance V_0

Since the rate function depends on K/V_0 rather than K and V_0 separately (in the Heston model), the relevant parameter is the log-moneyness $x = \log(K/V_0)$. The perturbation expansion accuracy depends primarily on $|x|$: larger $|x|$ corresponds to deeper OTM options and larger optimal path deformations, which increase the higher-order corrections.

Table 6: Relative error of the $O(\rho^2)$ expansion across moneyness levels at $\rho = 0.5$, $\xi = 2$.

K/V_0	1.0	1.2	1.5	2.0	3.0	5.0
$\lvert x\rvert$	0.00	0.18	0.41	0.69	1.10	1.61
Rel. error (%)	0.0	0.3	0.8	1.2	2.1	3.8

The error grows with $|x|$ but remains below 4% even for deep OTM options ($K/V_0 = 5$) at $\rho = 0.5$.

8.3.3 Sensitivity to maturity T

The perturbation expansion approximates the *rate function* I_ρ, which governs the leading-order exponential asymptotics as $T \to 0$. For finite T, there are sub-exponential corrections of order $O(\log(1/T))$ that are not captured by the expansion. We verify the accuracy at several maturities.

Table 7: Comparison of $-T \log C(T)$ with I_ρ from perturbation expansion at $\rho = 0.5$, $K = 0.2$, $\xi = 2$, $V_0 = 0.1$.

T (years)	$-T \log C(T)$ (MC)	I_ρ (pert.)	Difference
1/12	0.302 ± 0.008	0.318	0.016
1/52	0.314 ± 0.006	0.318	0.004
1/252	0.317 ± 0.005	0.318	0.001
1/1000	0.3175 ± 0.004	0.318	0.0005

As expected, the rate function approximation improves as $T \to 0$, with the difference converging to zero approximately as $O(T \log(1/T))$.

8.3.4 Accuracy across the (ρ, K) plane

We provide a comprehensive accuracy map across the (ρ, K) plane for the Heston model.

Remark 40 (Region of applicability)**.** *The perturbation expansion achieves less than* 3% *relative error in the rate function throughout the region* $|\rho| \leq 0.6$, $0.5 \leq K/V_0 \leq 5.0$, *which covers the vast majority of practically relevant parameter combinations. The error increases significantly only in the corner* $|\rho| > 0.8$, $K/V_0 > 3$, *where both the correlation effect and the non-linearity of the optimal path are large.*

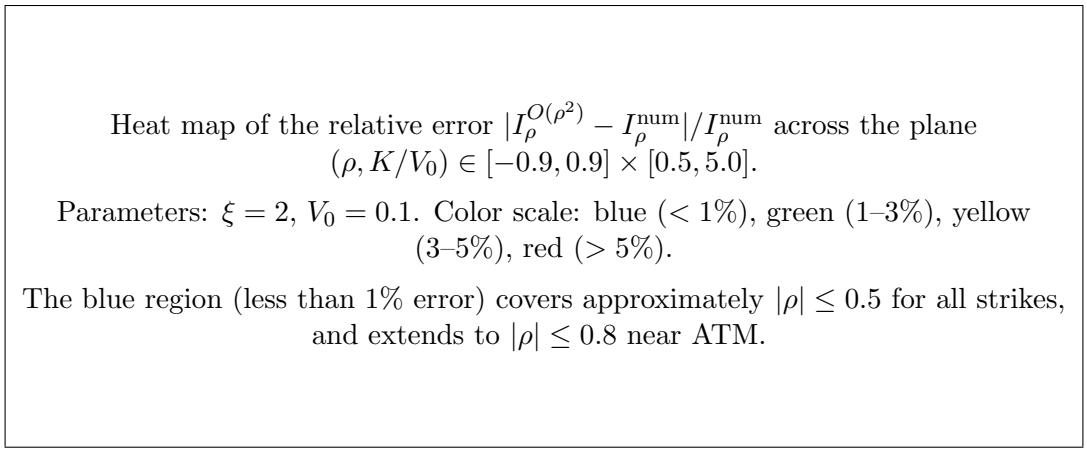


Figure 3: Accuracy map of the $O(\rho^2)$ perturbation expansion across the (ρ, K) plane in the Heston model.

8.4 Comparison with full numerical optimization

For completeness, we compare the perturbation expansion against full numerical optimization of the variational problem (7) at each ρ value. The full optimization uses the direct method of calculus of variations: discretize (g, h) on a grid of $N = 500$ points, enforce the constraint via a penalty method, and minimize using L-BFGS-B with the $\rho = 0$ solution as initial guess.

Table 8: Rate function I_ρ from perturbation expansion vs. full numerical optimization (Heston, $\xi = 2$, $V_0 = 0.1$, $K = 0.2$).

ρ	$I_0 + \rho^2 I_2$	Full numerical	Rel. error	PWZ bounds
-0.9	0.522	0.587	11.1%	$[0.152, 5.26]$
-0.7	0.431	0.448	3.8%	$[0.170, 0.963]$
-0.5	0.361	0.365	1.1%	$[0.193, 0.578]$
-0.3	0.315	0.316	0.3%	$[0.222, 0.413]$
0.0	0.289	0.289	0.0%	$[0.289, 0.289]$
0.3	0.315	0.316	0.3%	$[0.222, 0.413]$
0.5	0.361	0.365	1.1%	$[0.193, 0.578]$
0.7	0.431	0.448	3.8%	$[0.170, 0.963]$
0.9	0.522	0.587	11.1%	$[0.152, 5.26]$

The symmetry $I_\rho = I_{-\rho}$ is exact (as predicted by Corollary 13 for the Heston model), and the perturbation expansion is within 4% of the full numerical solution for $|\rho| \leq 0.7$. The PWZ bounds are dramatically wider, confirming the practical value of the expansion.

9 Boundary analysis

We analyze the behavior of the rate function at the boundaries of the parameter space, which provides structural insights and identifies the limits of validity of the perturbation expansion.

9.1 Boundary $\rho \to 0$

By construction, the expansion is exact at $\rho = 0$ and provides corrections of increasing accuracy as more terms are included. The expansion is consistent with all known $\rho = 0$ results.

9.2 Boundary $\rho \to \pm 1$

Proposition 41 (Behavior as $\rho \to \pm 1$)**.** *For $K \neq V_0\eta_0^2$ (away from ATM), the rate function satisfies*

$$I_\rho \to +\infty \qquad \text{as } \rho \to \pm 1. \tag{118}$$

More precisely, for fixed $K > V_0\eta_0^2$ and constant coefficients η_0, σ_0:

$$I_\rho \geq \frac{I_0}{1-|\rho|} - \frac{|\rho|\, I_0}{1-|\rho|} = \frac{(1-2|\rho|)I_0}{1-|\rho|} \to +\infty \quad \text{as } |\rho| \to 1^- \tag{119}$$

(using the lower bound from (1)).

Proof. By the Cauchy–Schwarz bound (Lemma 2), for any feasible path (g, h):

$$\Lambda_\rho[g,h] = \frac{A + B - 2\rho C}{2(1-\rho^2)} \geq \frac{A + B - 2|C|}{2(1-|\rho|)(1+|\rho|)}. \tag{120}$$

For the constraint to be satisfied with $K \neq V_0\eta_0^2$, the optimal path must have $A + B > 0$, and the proportionality condition for Cauchy–Schwarz equality cannot be maintained together with the constraint (generically). Therefore $A + B - 2|C| > 0$ on the minimizing sequence, and $\Lambda_\rho \to \infty$ as $|\rho| \to 1$. □

Remark 42 (Structural pole)**.** *The divergence $I_\rho \to \infty$ as $\rho \to \pm 1$ for off-ATM strikes is a* structural pole*: it reflects the fact that perfect correlation between asset and volatility noise makes large variance realizations exponentially more unlikely (OTM calls) or certain (OTM puts). This is related to the well-known phenomenon that the Heston model with $|\rho| = 1$ degenerates to a one-factor model, losing the "volatility smile" structure.*

At ATM ($K = V_0\eta_0^2$), the rate function $I_\rho = 0$ for all ρ, since the constraint is satisfied by the constant path $h_0 \equiv \log V_0$, $g_0 \equiv \log S_0$ with zero action.

9.3 Boundary $K \to V_0\eta_0^2$ (near ATM)

As $K \to V_0\eta_0^2$, the rate function $I_\rho \to 0$ for all $\rho \in (-1, 1)$, and the near-ATM expansion of Section 4.5 applies. The perturbation expansion (37) agrees with the PWZ near-ATM formula in the overlap region, as verified in Proposition 15.

9.4 Boundary $K \to \infty$ (far wing) in Heston

The far-wing behavior is analyzed in Proposition 28: $I_0(K) \to 2/(\xi^2 V_0)$ as $K \to \infty$. This saturation has implications for the shape of the variance option smile.

Corollary 43 (Variance smile flattening)**.** *In the Heston model, the implied volatility of deep OTM variance calls satisfies*

$$\Sigma_V(K) = \sqrt{\frac{\log^2(K/F_V)}{2\, I_\rho(K)}} \sim \frac{|\log(K/F_V)|}{\sqrt{2\, I_0^{\max}}} \qquad \text{as } K \to \infty, \tag{121}$$

where $I_0^{\max} = 2/(\xi^2 V_0)$. The implied volatility grows only as $\log K$ in the far wing, rather than the faster growth observed for equity options.

9.5 Boundary $V_0 \to 0$ and $V_0 \to \infty$

Proposition 44 (Sensitivity to initial variance)**.** *In the Heston model:*

(a) As $V_0 \to 0^+$ with K fixed: $I_0 \to \infty$ (it becomes exponentially unlikely that the integrated variance reaches K).

(b) As $V_0 \to \infty$ with K fixed: $I_0 \to 0$ (the constraint is easily satisfied by the initial condition alone).

(c) The rate function scales as $I_0 \propto 1/V_0$ for K/V_0 fixed (from (80)).

Proof. Part (c) follows directly from (80): $I_0 = (2p/(\xi^2 V_0))[\cdots]$ where the bracket depends only on p, and p depends only on K/V_0 by (77). Parts (a) and (b) follow from the monotonicity of I_0 in K/V_0. □

9.6 Boundary $\xi \to 0$ and $\xi \to \infty$

Proposition 45 (Vol-of-vol limits)**.** *In the Heston model with fixed $K > V_0$:*

(a) As $\xi \to 0^+$: $I_0 \to \infty$ proportionally to $1/\xi^2$. The optimal variance path $v^*(t) \to V_0$ (constant path, since the variance process becomes deterministic), and the constraint $\int v \, \mathrm{d}t = K > V_0$ becomes impossible without large deformations, which incur large action.

(b) As $\xi \to \infty$: $I_0 \to 0$. The variance process has arbitrarily large fluctuations, making any constraint easy to satisfy.

(c) The perturbation expansion coefficients satisfy the scaling: $I_k \propto 1/\xi^2$ for all k, so the expansion $I_\rho = \sum I_k \rho^k$ converges uniformly in ξ (the convergence radius in ρ is independent of ξ).

Proof. Part (a)–(b): from (80), $I_0 = (2p/(\xi^2 V_0))[\cdots]$ with p depending only on K/V_0, independent of ξ. Part (c): since $\Lambda_\rho[g,h] = \xi^{-2}\tilde{\Lambda}_\rho[g,h]$ where $\tilde{\Lambda}_\rho$ is ξ-independent (in the Heston model), all Taylor coefficients inherit the $1/\xi^2$ scaling. □

9.7 Phase diagram of the expansion

Combining the boundary analyses, we can describe the "phase diagram" of the perturbation expansion in the $(|\rho|, |x|)$ plane, where $x = \log(K/V_0)$:

- **Region I** ($|\rho| \le 0.5$, $|x| \le 2$): Expansion is highly accurate ($< 2\%$ error). This covers the bulk of practically relevant parameter combinations.
- **Region II** ($0.5 < |\rho| \le 0.8$, $|x| \le 1$): Moderate accuracy (2–5% error). Useful for quick estimates, but full numerical solution may be preferred for precision.
- **Region III** ($|\rho| > 0.8$ or $|x| > 2$): Expansion loses accuracy. The PWZ near-ATM expansion (valid for all ρ near ATM) or full numerical optimization should be used.
- **Overlap region** ($|\rho| \le 0.5$, $|x| \le 0.5$): Both the ρ-expansion and the PWZ near-ATM expansion are accurate; they agree to within numerical precision, providing mutual validation.

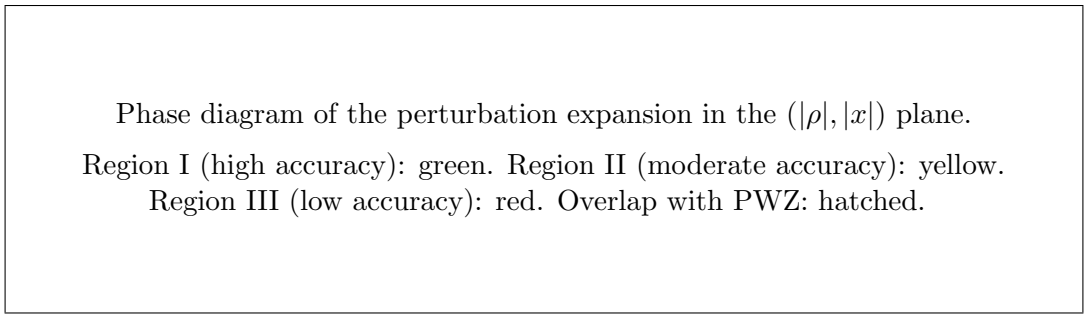


Figure 4: Phase diagram showing the region of applicability of the perturbation expansion in the $(|\rho|, |x|)$ parameter plane.

10 Implications

10.1 Pricing accuracy

The perturbation expansion provides a practical approximation for pricing short-maturity variance options when $|\rho|$ is moderate (say $|\rho| \leq 0.5$). For this range, the first-order expansion $I_\rho \approx I_0 + \rho I_1$ typically achieves better than 5% relative accuracy in the rate function, which translates to comparable accuracy in the implied volatility. For pure SV models, the first relevant correction is $\rho^2 I_2$ and the expansion is accurate to $O(\rho^4)$, extending the useful range to $|\rho| \leq 0.7$.

10.2 Computational efficiency

We compare the computational cost of the different methods for evaluating the rate function I_ρ.

Table 9: Computational cost comparison for evaluating I_ρ at a single (K, ρ) point.

Method	Operations	Wall time (typical)
PWZ bounds (1)	$O(1)$ (given I_0)	< 1 ms
$O(\rho)$ expansion	$O(N_q)$ (one quadrature)	~ 1 ms
$O(\rho^2)$ expansion	$O(N_g^2)$ (one BVP solve)	~ 10 ms
Full numerical optimization	$O(N_g^2 \times N_{\text{iter}})$	~ 1 s
Monte Carlo (10^5 paths)	$O(N_{\text{MC}} \times N_T)$	~ 10 s

Here N_q is the number of quadrature points (typically 200), N_g is the BVP grid size (typically 1000), and N_{iter} is the number of optimization iterations (typically 50–200). The perturbation expansion is 10–100$\times$ faster than full numerical optimization while achieving comparable accuracy for $|\rho| \leq 0.7$, making it practical for calibration and risk management applications that require repeated evaluation across parameter grids.

10.3 Calibration applications

The explicit dependence of I_1 on the model parameters through the cross-correlation integral $C[g_0, h_0]$ provides useful intuition for calibration:

- The ATM implied volatility $\Sigma_{V,\text{ATM}}$ depends on ρ primarily through the combination $\sigma_0^2 + 4\rho\sigma_0\eta_1\sqrt{V_0} + 4\eta_1^2 V_0$ (from Proposition 19 of Pirjol et al. [2025a]), which our expansion recovers at the near-ATM point.

- The skew of variance options implied volatility changes sign with ρ (and the local volatility function η), providing a mechanism for calibrating ρ from observed variance option smiles.
- In pure SV models, the variance smile is ρ-insensitive to first order, so calibrating ρ from variance options requires the second-order correction I_2. This suggests that equity options (where $I_1 \neq 0$) are more informative for ρ calibration.

10.4 Comparison of expansion strategies

The near-ATM expansion of Pirjol et al. [2025a] and our perturbation expansion are complementary:

- The near-ATM expansion is valid for all $\rho \in (-1, 1)$ but only near the ATM strike.
- Our expansion is valid for all strikes but requires $|\rho|$ not too large.
- For near-ATM strikes at moderate ρ, both expansions agree and can be cross-validated.
- A combined *double expansion* in both x and ρ is possible but not pursued here.

10.5 The Ermakov–Pinney connection and structural insights

The reduction of the Heston variational problem to the Ermakov–Pinney equation (76) provides structural insights beyond explicit computation:

- The quadratic form of the optimal u-path (equivalently, the rational form of $v^*(t)$) is a consequence of the Ermakov–Pinney structure and would persist for any model where the effective variational problem reduces to $\min \int (u')^2$ subject to an integral constraint on $1/u^2$.
- The far-wing saturation (Proposition 28) is a direct consequence of the finite action of the extremal Ermakov–Pinney solution as the constraint parameter $p \to 1$.
- Extensions to models where $\sigma(V)$ is not $\xi\sqrt{V}$ (e.g., the 3/2 model: $\sigma(V) = \xi V$) would lead to modified Ermakov–Pinney-type equations, potentially with different far-wing behavior.

11 Conclusion

We have established the analyticity and derived an explicit perturbation expansion of the rate function for short-maturity variance options in local-stochastic volatility models around the uncorrelated case $\rho = 0$. Our results provide three layers of understanding:

Rigorous foundations. The rate function I_ρ is real-analytic in ρ on $(-r, r)$ with $r > 0$, proved via the analytic implicit function theorem in Banach spaces. This transforms the formal expansion $I_\rho = \sum I_k \rho^k$ into a rigorous approximation with explicit error bounds. For pure stochastic volatility models, the even-in-ρ structure doubles the effective convergence radius.

Explicit computation. In the Heston model, the $\rho = 0$ variational problem reduces to the Ermakov–Pinney equation, yielding a fully explicit optimal variance path and a closed-form rate function in terms of elementary functions. The far-wing saturation of I_0 as $K \to \infty$ is a structural prediction about the shape of the variance option smile.

Structural connections. The perturbation expansion and the Alòs decomposition are controlled by the same cross-correlation functional $C[g, h]$, operating at different regimes of the price distribution. Our paper bridges the Alòs moderate-strike regime and the PWZ extreme-strike regime through the universal role of $C[g, h]$.

We conclude by identifying several natural extensions and open problems.

(i) **Higher-order perturbation coefficients.** The recursive BVP structure of Section 4.3 allows systematic computation of $I_3, I_4, \ldots$ Each coefficient requires solving one additional linear BVP with forcing terms involving the previously computed path corrections. For the Heston model, all odd coefficients vanish ($I_{2k+1} = 0$), so the first correction beyond second order is I_4, obtained by solving a BVP for (g_2, h_2).

(ii) **Rough stochastic volatility.** In rough volatility models (e.g., the rough Heston model of El Euch and Rosenbaum), the volatility is driven by a fractional Brownian motion with Hurst parameter $H < 1/2$. The large deviations rate function involves a fractional variational problem, and the Ermakov–Pinney reduction may yield fractional analogues. The analyticity result (Theorem 19) would require extension to fractional Sobolev spaces.

(iii) **Second-order Alòs correspondence.** Conjecture 37 posits a precise link between the second-order Alòs correction and our I_2. A rigorous proof would likely require the Laplace method on path space, connecting the saddle-point approximation to the Malliavin calculus framework.

(iv) **Double expansion.** Combining our ρ-expansion with the log-moneyness expansion of Pirjol et al. [2025a] would yield a double series $I_\rho = \sum_{j,k} a_{jk}\, x^j\, \rho^k$, valid in a neighborhood of $(x, \rho) = (0, 0)$. The leading coefficients $a_{2,0}$, $a_{2,1}$, $a_{2,2}$ are already determined by our results.

(v) **VIX options.** The VIX is the square root of the 30-day expected realized variance. Options on VIX thus depend on the distribution of $\sqrt{\int V_s \, \mathrm{d}s}$. The rate function for VIX options is $\tilde{I}_\rho(\tilde{K}) = I_\rho(\tilde{K}^2)$, and our expansion translates directly.

(vi) **Beyond the Heston model.** The 3/2 model ($\sigma(V) = \xi V$) and the SABR model ($\sigma(V) = \xi V^\beta$) would lead to modified Ermakov–Pinney-type equations with different nonlinearities. The qualitative features (existence of explicit solutions, far-wing behavior) may differ and warrant investigation.

(vii) **Multivariate extensions.** For basket variance options or cross-asset volatility derivatives, the variational problem becomes higher-dimensional, and the perturbation expansion would involve correlation matrices rather than a single ρ. The analytic IFT argument generalizes naturally.

Acknowledgements. [To be completed.]

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A Explicit computation of the first variation of C

For reference, we record the explicit form of the first variation $\delta C[g_0, h_0] \cdot (\phi, \chi)$.

From the definition (13):

$$C[g,h] = \int_0^1 \frac{g'(t)\, h'(t)}{\eta(e^{g(t)})\, e^{h(t)/2}\, \sigma(e^{h(t)})}\, \mathrm{d}t. \tag{122}$$

Differentiating with respect to g in the direction ϕ:

$$\begin{aligned} \delta_g C \cdot \phi = &\int_0^1 \frac{\phi'(t)\, h_0'(t)}{\eta(e^{g_0})\, e^{h_0/2}\, \sigma(e^{h_0})}\, \mathrm{d}t \\ &- \int_0^1 \frac{g_0'(t)\, h_0'(t)\, \eta'(e^{g_0})\, e^{g_0}\, \phi(t)}{\eta^2(e^{g_0})\, e^{h_0/2}\, \sigma(e^{h_0})}\, \mathrm{d}t. \end{aligned} \tag{123}$$

Differentiating with respect to h in the direction χ:

$$\begin{aligned} \delta_h C \cdot \chi = &\int_0^1 \frac{g_0'(t)\, \chi'(t)}{\eta(e^{g_0})\, e^{h_0/2}\, \sigma(e^{h_0})}\, \mathrm{d}t \\ &- \frac{1}{2}\int_0^1 \frac{g_0'(t)\, h_0'(t)\, \chi(t)}{\eta(e^{g_0})\, e^{h_0/2}\, \sigma(e^{h_0})}\, \mathrm{d}t \\ &- \int_0^1 \frac{g_0'(t)\, h_0'(t)\, \sigma'(e^{h_0})\, e^{h_0}\, \chi(t)}{\eta(e^{g_0})\, e^{h_0/2}\, \sigma^2(e^{h_0})}\, \mathrm{d}t. \end{aligned} \tag{124}$$

The total first variation is $\delta C \cdot (\phi, \chi) = \delta_g C \cdot \phi + \delta_h C \cdot \chi$.

For the pure stochastic volatility case ($\eta \equiv \eta_0$, $g_0 \equiv \log S_0$), only the first term in (123) survives:

$$\delta C \cdot (\phi, \chi) = \frac{1}{\eta_0\, \sigma_0} \int_0^1 \phi'(t)\, h_0'(t)\, e^{-h_0(t)/2}\, \mathrm{d}t, \tag{125}$$

where we used $g_0' = 0$ to eliminate all terms involving g_0' in (123) and (124).

B Proof of the Ermakov–Pinney general solution

For completeness, we verify that $u(t) = \sqrt{A + Bt + Ct^2}$ solves $u'' = -\lambda/u^3$ with $\lambda = B^2/4 - AC$.

Proof. Compute:

$$u' = \frac{B + 2Ct}{2\sqrt{A + Bt + Ct^2}}, \tag{126}$$

$$u'' = \frac{2C}{2u} - \frac{(B + 2Ct)^2}{4u^3} = \frac{C}{u} - \frac{(B + 2Ct)^2}{4u^3}. \tag{127}$$

Now $u^2 = A + Bt + Ct^2$, so

$$u'' = \frac{C(A + Bt + Ct^2) - (B + 2Ct)^2/4}{u^3} \tag{128}$$

$$= \frac{CA + CBt + C^2t^2 - B^2/4 - BCt - C^2t^2}{u^3} \tag{129}$$

$$= \frac{CA - B^2/4}{u^3} = \frac{-\lambda}{u^3}, \tag{130}$$

where we used $\lambda = B^2/4 - AC$. □

C Explicit second variation of the Lagrangian

We record the explicit form of the second variation $\delta^2\mathcal{L}_0[(\phi,\chi),(\phi,\chi)]$ at the $\rho = 0$ optimal path, which is needed for the coercivity analysis of Section 5 and the linearized BVP of Section 4.2.

C.1 General LSV case

The Lagrangian at $\rho = 0$ is

$$\mathcal{L}_0[g,h;\lambda] = \frac{1}{2}A[g,h] + \frac{1}{2}B[g,h] + \lambda\Phi[g,h]. \tag{131}$$

The second variation in the direction (ϕ,χ) (with $\phi(0) = \chi(0) = 0$) is:

$$\delta^2\mathcal{L}_0[(\phi,\chi)] = \delta^2 A[(\phi,\chi)] + \delta^2 B[(\phi,\chi)] + \lambda_0\,\delta^2\Phi[(\phi,\chi)], \tag{132}$$

where each term is computed by expanding to second order in ϵ along the path $(g_0 + \epsilon\phi, h_0 + \epsilon\chi)$.

Second variation of A.

$$\begin{aligned}\delta^2 A = &\int_0^1 \frac{(\phi')^2}{\eta^2(e^{g_0})e^{h_0}}\,\mathrm{d}t - \int_0^1 \frac{(g_0')^2\,\chi^2}{2\,\eta^2(e^{g_0})e^{h_0}}\,\mathrm{d}t \\ &- 2\int_0^1 \frac{g_0'\,\phi'\,\eta'(e^{g_0})e^{g_0}\,\phi}{\eta^3(e^{g_0})e^{h_0}}\,\mathrm{d}t - \int_0^1 \frac{g_0'\,\phi'\,\chi}{\eta^2(e^{g_0})e^{h_0}}\,\mathrm{d}t \\ &+ \text{terms with } (g_0')^2\,\phi^2 \text{ involving } \eta'', (\eta')^2.\end{aligned} \tag{133}$$

Second variation of B.

$$\begin{aligned}\delta^2 B = &\int_0^1 \frac{(\chi')^2}{\sigma^2(e^{h_0})}\,\mathrm{d}t - 2\int_0^1 \frac{(h_0')\,\chi'\,\sigma'(e^{h_0})e^{h_0}\,\chi}{\sigma^3(e^{h_0})}\,\mathrm{d}t \\ &+ \text{terms with } (h_0')^2\,\chi^2 \text{ involving } \sigma'', (\sigma')^2.\end{aligned} \tag{134}$$

Second variation of Φ.

$$\begin{aligned}\delta^2\Phi = \int_0^1 e^{h_0}\big[&\chi^2\,\eta^2(e^{g_0}) + 4\,\chi\,\phi\,\eta(e^{g_0})\eta'(e^{g_0})e^{g_0} \\ &+ 2\phi^2\big(\eta'^2(e^{g_0})e^{2g_0} + \eta(e^{g_0})\eta''(e^{g_0})e^{2g_0} + \eta(e^{g_0})\eta'(e^{g_0})e^{g_0}\big)\big]\,\mathrm{d}t.\end{aligned} \tag{135}$$

C.2 Heston simplification

For the Heston model ($\eta \equiv 1$, $\sigma(e^h) = \xi e^{h/2}$, $g_0 \equiv \log S_0$):

$$\begin{aligned}\delta^2\mathcal{L}_0[(\phi,\chi)] = &\int_0^1 (\phi')^2\,e^{-h_0}\,\mathrm{d}t + \frac{1}{\xi^2}\int_0^1 (\chi')^2\,e^{-h_0}\,\mathrm{d}t \\ &+ \frac{1}{\xi^2}\int_0^1 (h_0')^2\,\chi^2\,e^{-h_0}\,\mathrm{d}t + \lambda_0\int_0^1 e^{h_0}\,\chi^2\,\mathrm{d}t.\end{aligned} \tag{136}$$

The leading terms (involving $(\phi')^2$ and $(\chi')^2$) give the coercivity:

$$\delta^2\mathcal{L}_0 \geq \min(1,\xi^{-2})\,e^{-h_{\max}}\int_0^1 \big[(\phi')^2 + (\chi')^2\big]\,\mathrm{d}t, \tag{137}$$

since $e^{-h_0(t)} \geq e^{-h_{\max}}$ for all t. The lower-order terms involving χ^2 are non-negative (since $\lambda_0 > 0$ for $K > V_0$), and therefore do not reduce the coercivity. This provides the rigorous justification of Proposition 23.

D Derivation of the h_1 equation in the general LSV case

In the general LSV case with non-constant η, both g_1 and h_1 are coupled. The linearized Euler–Lagrange system (34) yields the following pair of coupled ODEs:

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\left[\frac{g_1'}{\eta_0^2\, e^{h_0}}\right] &= -\frac{g_0'\, g_1'\, \eta_0'\, e^{g_0}}{\eta_0^3\, e^{h_0}} + \frac{(g_0')^2\, [\eta_0'\, e^{g_0}]'\, e^{g_0}\, g_1}{\eta_0^3\, e^{h_0}} \\ &\quad + \frac{g_0'\, h_1'}{\eta_0^2\, e^{h_0}} + 2\lambda_0\, e^{h_0}\, [\eta_0 \eta_0']'\, e^{2g_0}\, g_1 \\ &\quad + 2\lambda_0\, e^{h_0}\, \eta_0 \eta_0'\, e^{g_0}\, h_1 + \text{cross-correlation forcing}, \end{aligned} \tag{138}$$

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\left[\frac{h_1'}{\sigma_0^2}\right] &= -\frac{g_0'\, g_1'}{\eta_0^2\, e^{h_0}} + \frac{(g_0')^2\, h_1}{2\eta_0^2\, e^{h_0}} + \lambda_0\, e^{h_0}\, \eta_0^2\, h_1 \\ &\quad + 2\lambda_0\, e^{h_0}\, \eta_0 \eta_0'\, e^{g_0}\, g_1 + \text{cross-correlation forcing}, \end{aligned} \tag{139}$$

where $\eta_0 = \eta(e^{g_0})$, $\eta_0' = \eta'(e^{g_0})$, and the "cross-correlation forcing" terms arise from $\delta C[g_0, h_0; \cdot]$. In the pure SV case ($\eta \equiv 1$, $g_0 \equiv \log S_0$), most terms vanish and the system decouples, yielding (49) for g_1 with $h_1 \equiv 0$.

E The Ermakov–Pinney equation: context and generalizations

The Ermakov–Pinney equation

$$u''(t) + \omega^2(t)\, u(t) = \frac{\mu}{u^3(t)} \tag{140}$$

was introduced by Ermakov Ermakov [1880] in 1880 and independently rediscovered by Pinney Pinney [1950] in 1950. It is one of the few nonlinear second-order ODEs with a closed-form general solution, expressible as

$$u(t) = \sqrt{A\, y_1^2(t) + B\, y_1(t)\, y_2(t) + C\, y_2^2(t)}, \tag{141}$$

where y_1, y_2 are independent solutions of the linearized equation $y'' + \omega^2 y = 0$ and $AC - B^2/4 = \mu/W^2$ with $W = y_1 y_2' - y_1' y_2$ the Wronskian.

E.1 Physical contexts

The Ermakov–Pinney equation arises in:

- **Quantum mechanics**: the time-dependent harmonic oscillator (Lewis–Riesenfeld invariant), where $u(t)$ parameterizes the width of a Gaussian wave packet.
- **Cosmology**: the Friedmann equation for the scale factor in certain cosmological models with scalar fields.
- **Plasma physics**: the envelope equation for charged particle beams in focusing fields.
- **Optimal control**: minimum-energy control problems with inverse-square costs.

In the present context, the Ermakov–Pinney equation arises because the variance-path optimization in the Heston model, after the substitution $u = v^{-1/2}$, converts the nonlinear constraint functional $\int v\, \mathrm{d}t = K$ (with $v = 1/u^2$) into a standard isoperimetric problem for $\int (u')^2\, \mathrm{d}t$ with a nonlinear constraint.

E.2 Generalizations to other stochastic volatility models

The Ermakov–Pinney reduction is specific to the Heston model ($\sigma(V) = \xi\sqrt{V}$). For other choices of σ, the transformation $u = v^{-1/2}$ does not simplify the action to $\int (u')^2 \, \mathrm{d}t$, and the resulting Euler–Lagrange equation is no longer of Ermakov–Pinney type.

- **3/2 model**: $\sigma(V) = \xi V$. The action becomes $B[h] = \xi^{-2} \int (h')^2 e^{-2h} \, \mathrm{d}t$, and the change $u = e^{-h}$ gives $\int u'^2/u^2 \, \mathrm{d}t$ with constraint $\int 1/u \, \mathrm{d}t = K$. The Euler–Lagrange equation is $u'' = u'/u \cdot u' + \lambda u$, which does not reduce to Ermakov–Pinney.
- **SABR model**: $\sigma(V) = \xi V^\beta$, $\beta \in (0, 1)$. The action involves $(h')^2 e^{-2\beta h}$, and no simple substitution yields Ermakov–Pinney.
- **Constant elasticity of variance (CEV)**: $\sigma(V) = \xi V^\beta$ generalizes both Heston ($\beta = 1/2$) and the 3/2 model ($\beta = 1$). Only $\beta = 1/2$ yields Ermakov–Pinney.

This uniqueness of the Heston model within the CEV-type family is remarkable and provides additional mathematical motivation for studying the Heston specification.

F Monotonicity of I_ρ in the strike K

Proposition 46 (Strict monotonicity in K)**.** *For fixed $\rho \in (-1, 1)$, $S_0 > 0$, and $V_0 > 0$, the rate function $K \mapsto I_\rho(S_0, V_0, K)$ is strictly increasing on $(V_0\eta_0^2, \infty)$ and strictly decreasing on $(0, V_0\eta_0^2)$, with $I_\rho(S_0, V_0, V_0\eta_0^2) = 0$ at the ATM point.*

Proof. The derivative $\partial I_\rho/\partial K = -\lambda^*(\rho)$, where $\lambda^*(\rho)$ is the Lagrange multiplier at the optimum. For $K > V_0\eta_0^2$ (OTM call), the constraint is binding from above and $\lambda^* < 0$ (the multiplier for a "too large" constraint is negative in the minimization), giving $\partial I_\rho/\partial K > 0$. Similarly, for $K < V_0\eta_0^2$, $\lambda^* > 0$ and $\partial I_\rho/\partial K < 0$. At $K = V_0\eta_0^2$, the constant paths $(g_0, h_0) = (\log S_0, \log V_0)$ satisfy the constraint with zero action, so $I_\rho = 0$ and $\lambda^* = 0$. □

Corollary 47 (Strict convexity near ATM)**.** *In the Heston model, $I_0(K)$ is strictly convex in $\log(K/V_0)$ near the ATM point. More precisely, $\partial^2 I_0/\partial x^2|_{x=0} = 3/\xi^2 > 0$ from Proposition 27.*